

HYDRAULIC SIMULATION FOR OTTER FIELD DEVELOPMENT

Report

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1 INTRODUCTION

The purpose of this report is to present the hydraulic analysis for the Otter field development subsea production control system. This simulation is based on tender specifications and is to be understood as a preliminary report.

The analysis addresses pressure levels and design features required for operating the system according to requirements set by the customer.

The XTs are equipped with the following valves:

- 3 off 5 1/8" FMC Gate Valves Model 130 (PWV, PMV and MIV)
- 2 off 2 1/16" FMC Gate Valves Model 130 (XOV and AMV)
- 2 off 1" FMC Gate Valves Model 140 (PIV and CPIV)
- 1 off SCSSV – Baker 5" (chosen for simulation purpose)

Simulation Scenario:

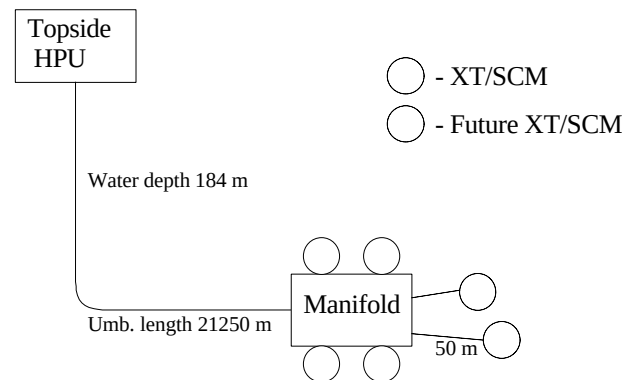


Figure 1-1 Simulation scenario

The report contains the following main items:

- Conclusion
- Description of analysis method
- Configurations to be analyzed.
- Input data
- Results

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1.1 REFERENCES

/1/ TNO-0010410	Verification of the Hydraulic Simulation Program
/2/ OTT-U10-PT-80-0411	Functional Specification – Subsea Production Data Monitoring System
(TOTALFINA Doc.)	
/3/ SPC-0000020339	Functional Specification Otter Umbilical, Rev. B

1.2 ABBREVIATIONS

ESD	- Emergency Shut Down
HP	- High Pressure
HPU	- Hydraulic Power Unit
ID	- Inner Diameter
LP	- Low Pressure
SCM	- Subsea Control Module
SCSSV	- Surface Controlled Sub-surface Safety Valve
XT	- X-mas Tree

2 CONCLUSION

The results shows that the LP-system will function satisfactory and meet all the design requirements with 12.7 mm ID flexible supply lines, opening flow restrictor of 3480 bar/(l/s)² and without any subsea accumulators. As per company request implementation of subsea accumulators is performed. This has been simulated in chosen cases.

The HP system will have a longer umbilical charge-up time than specified in the design requirements, else is the system within the requirements. It is recommended to reduce HP umbilical to 3/8" ID and increase the HPU capacity from 2.8 l/min to 4.2 l/min by add on one more HP pump in parallel. The simulations show that these modifications will be enough to meet the design requirements.

Note: The recommendations mentioned above are implemented.

The Choke operation time is 7 minutes from fully opened to fully closed position and vice versa.

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3 STUDY METHODOLOGY

The hydraulic analysis has been carried out using a computer simulation program. The program establishes a number of first order non-linear differential equations, one equation per descriptive variable. The set of equations is solved using a variable time-step Runge Kutta algorithm.

The following physical effects are included in the mathematical model:

- Pressure losses due to wall friction in long conduits.
- Turbulent / laminar flow regime
- Pressure losses due to changes in fluid velocity in valves, couplers, flow bends and changes in cross sectional area etc.
- Fluid compressibility.
- Fluid viscosity as a function of actual pressure and temperature.
- Fluid density as a function of actual pressure and temperature.
- Water depth of deployed units.
- HPU height above sea level.
- Real gas behavior of nitrogen (Bender-Benedict-Webb-Rubin).
- Actuator friction as a function of hydraulic fluid pressure.

For describing hydraulic restrictions, a unit denoted Rh, which differs somewhat from the traditional Cv, is used within this document.

Rh is defined as follows:

$$Rh = \frac{\Delta P \cdot 870}{Q^2 \cdot \rho} \quad (\text{bar} / (\text{l/s})^2)$$

Here, ΔP = Pressure drop across restriction (bar)
 Q = Flow rate through the restriction (l/s)
 ρ = Actual density of fluid (kg/m³)

At a flow rate of 0.1 l/s, the pressure drop with HW 540 through a restriction of 430 bar/(l/s)² will then be:

$$\Delta P = 430 \cdot 0.1 \cdot 0.1 \cdot 1057/870 = 5.2 \text{ bar}$$

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4 DESIGN REQUIREMENTS

Customer design requirements according to ref. /2/:

- Interval between Xmas Tree start-ups: 15 min.
- Xmas Tree single valve opening times: < 1 min.
- Xmas Tree single valve closing times: < 45 sec.
- SCSSV opening times. 15 sec.
- SCSSV closing times. 45 sec.
- Manifold single valve opening times: < 1 min.
- Manifold single valve closing times: < 45 sec.
- ESD shutdown time – controlled: < 5 min.
- ESD shutdown time – hydraulic vent down: < 45 min.
- Charge up of umbilical: < 60 min.
- Choke valve opening time: Contractor

FMC – system design requirements:

- Maintain the pressure difference between SCM supply and return pressures 15% above the highest start-to-close pressure during opening of an actuator.
- Maintain the pressure difference between SCM supply and return pressures 50% above the control valve reset pressure (65 bar) during opening of an actuator.

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5 CONFIGURATIONS TO BE ANALYZED

Case:	Description
1a	Operating a 5" valve with a HPU outlet pressure of 190 bar. Well head pressure of 89 bar. Flowline pressure of 1 bar. Opening flow restrictions of 3450 bar/(l/s) ² .
1b	Operating a 2" valve with a HPU outlet pressure of 190 bar. Well head pressure of 89 bar. Flowline pressure of 1 bar. Opening flow restrictions of 3480 bar/(l/s) ² .
1c	Operating a 1" valve with a HPU outlet pressure of 190 bar. Well head pressure of 89 bar. Flowline pressure of 1 bar. Opening flow restrictions of 3480 bar/(l/s) ² .
2a sequence.	Opening a XT, two 5" valves, two 2" valves and two 1" valves in HPU outlet pressure of 190 bar. Well head pressure of 89 bar. Flowline pressure of 89 bar. Opening flow restrictions of 3480 bar/(l/s) ² .
2b sequence.	Closing a XT, two 5" valves, two 2" valves and two 1" valves in HPU outlet pressure of 190 bar. Well head pressure of 89 bar. Flowline pressure of 89 bar. Opening flow restrictions of 3480 bar/(l/s) ² .
3	Choke operation, Masterflo-P4. HPU outlet pressure of 190 bar. Well head pressure of 89 bar. Flowline pressure of 89 bar. Opening flow restrictions of 3480 bar/(l/s) ² .
4a	Charging up the LP umbilical line.
4b	Bleeding down the LP umbilical line.
5	Operating a DHSV HPU outlet pressure of 500 bar. Well head pressure of 89 bar. Flowline pressure of 1 bar.

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- 6a Charging up the HP umbilical.
- 6b Bleeding down the HP umbilical.
- 7a Operating a 5" valve with HPU outlet pressure of 190 bar.
Well head pressure of 89 bar.
Flowline pressure of 1 bar.
Opening flow restrictions of 3450 bar/(l/s)².
With 20 liter subsea accumulator on manifold.
- 7b Opening a XT, two 5" valves, two 2" valves and two 1" valves in sequence.
HPU outlet pressure of 190 bar.
Well head pressure of 89 bar.
Flowline pressure of 89 bar.
Opening flow restrictions of 3480 bar/(l/s)².
With 20 liter subsea accumulator on manifold.
- 7c Choke operation, Masterflo-P4.
HPU outlet pressure of 190 bar.
Well head pressure of 89 bar.
Flowline pressure of 89 bar.
Opening flow restrictions of 3480 bar/(l/s)².
With 20 liter subsea accumulator on manifold.
- 7d Charging up the LP umbilical line.
With 20 liter subsea accumulator on manifold.
- 7e Bleeding down the LP umbilical line.
With 20 liter subsea accumulator on manifold.

6 INPUT DATA

The following input data has been used:

Case	Brief description of case parameters	Value	Unit
General	Fluid:	Oceanic	HW540
	Fluid temperature	5	°C
	Viscosity at 5°C	10	cSt
	Density at 5°C	1040	kg/m ³
	Bulk modulus	24000	bar
	Water depth	184	m
	Distance from water surface to HPU	40	m
	Well head pressure	89	bar
	HPU pump displacement, LP	2 x 6	l/min
	HPU pump displacement, LP	2 x 1.4	l/min
	HPU outlet low-pressure	207	bar
	HPU outlet high-pressure	517	bar
	Umbilical length	21250	m
	Hydraulic supply line ID, LP/HP system	12.7	mm
	Control valve restriction (Rh)	3480	bar/(l/s) ²
	FMC M130-5 1/8" actuator, actuator volume	8.0	liter
	FMC M130-2 1/16" actuator, actuator volume	1.0	liter
	FMC M130-1" actuator, actuator volume	0.5	liter
	Masterflo P4 balanced choke, actuator volume	0.02075	liter
	Down hole valve type: Baker 5", actuator volume	22.2	cm ³
	5 1/8" manifold valve, actuator volume	2.3	liter

NOTE:

In order to compensate for hysteresis in the HPU outlet pressure regulating circuit, the HPU outlet pressure in all cases have been set to 190 bar for LP supply and 500 bar in HP supply.

7 RESULTS

7.1 XT VALVES CHARACTERISTICS

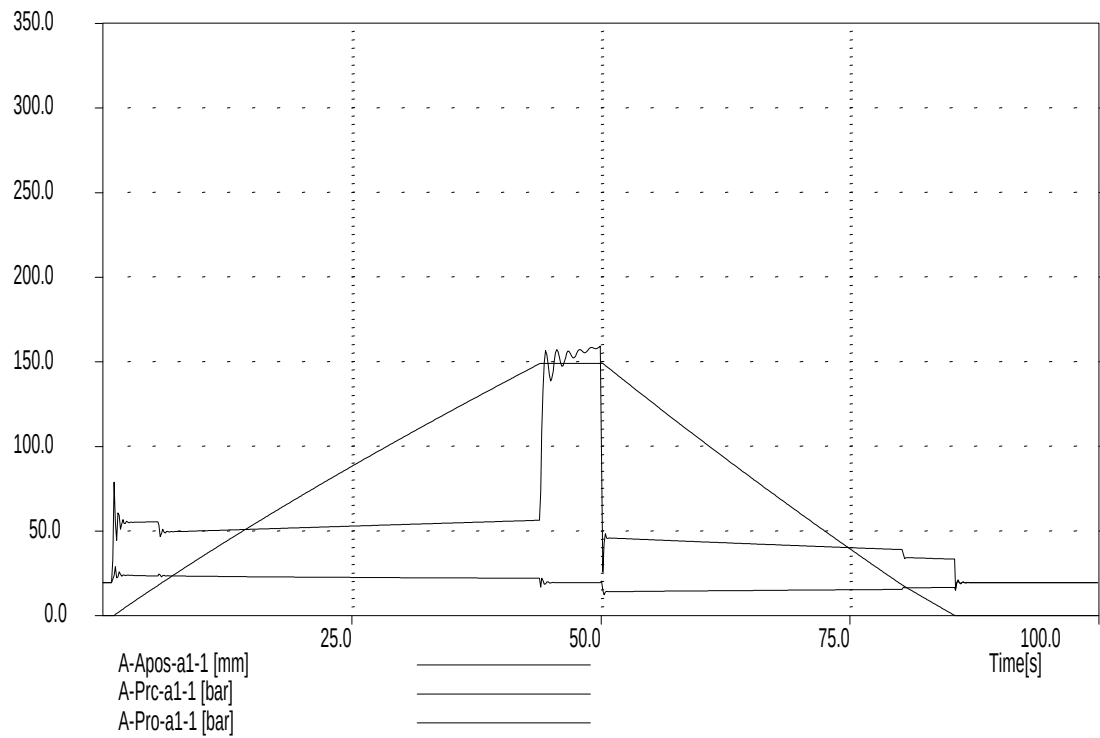
7.1.1 Case 1a, Operating a M130 - 5" valve

The characteristics of the M130 - 5" 690 bar gate valve is illustrated here. The first plot shows the opening and closing cavity pressures and the actuator position during operation of the valve. The open stroke takes 43 seconds (between $t = 1$ and 44 seconds) and the closing stroke takes 36 seconds (between $t = 50$ and 86 seconds). This opening and closing times is within the design requirements stated in chapter 4 of this document.

During the opening stroke the friction is at maximum before the pinch open point (gate bore connects to presses bore) is reached at $t = 5$ seconds. At this point the pressure across the gate is equalized and thus the friction is reduced as is reflected in a reduction in the opening cavity pressure (and increase in the return cavity pressure). Reaching the end position at $t = 44$ seconds, the opening cavity pressure increases to SCM actual supply pressure.

The opening cavity pressure during the closing stroke is less than during the opening stroke due to the spring force working to close the actuator. At $t = 80$ seconds the pinch point is reached and the friction across the gate increases as the gate pressure drop is increased. The effect of this is a drop in actuator opening cavity pressure and actuator velocity.

Observe that normal operation of valves will be with equal pressure on both sides of the gate (no pressure drop over the gate) and thus, minimized gate friction.

**Figure 7-1 Operation of a M130 - 5" gate valve**

Legend: A-Apos-a1-1; Actuator position (mm)
A-Prc-a1-1; Actuator closing cavity pressure (bar)
A-Pro-a1-1; Actuator opening cavity pressure (bar)

The actuator opening cavity pressure and the SCM supply and return pressures are visualized in the plot below.

The SCM supply pressure drops from 208 bar to some 127 bar during stroking the 8-liter actuator. This is well above the reset pressure of the actuator, which is approx. 35 bar according to figure 7-1.

The SCM return pressure will stabilize at approx. 21 bar during the opening of the valve and the minimum pressure difference between the SCM supply and return pressure is $127 - 21 = 106$ bar. This is above the reset pressure for the control valves which is $60 \times 1.5 = 97.5$ bar according to the design requirements. This means that the system is able to open a 5" actuator without assistance from a subsea accumulator.

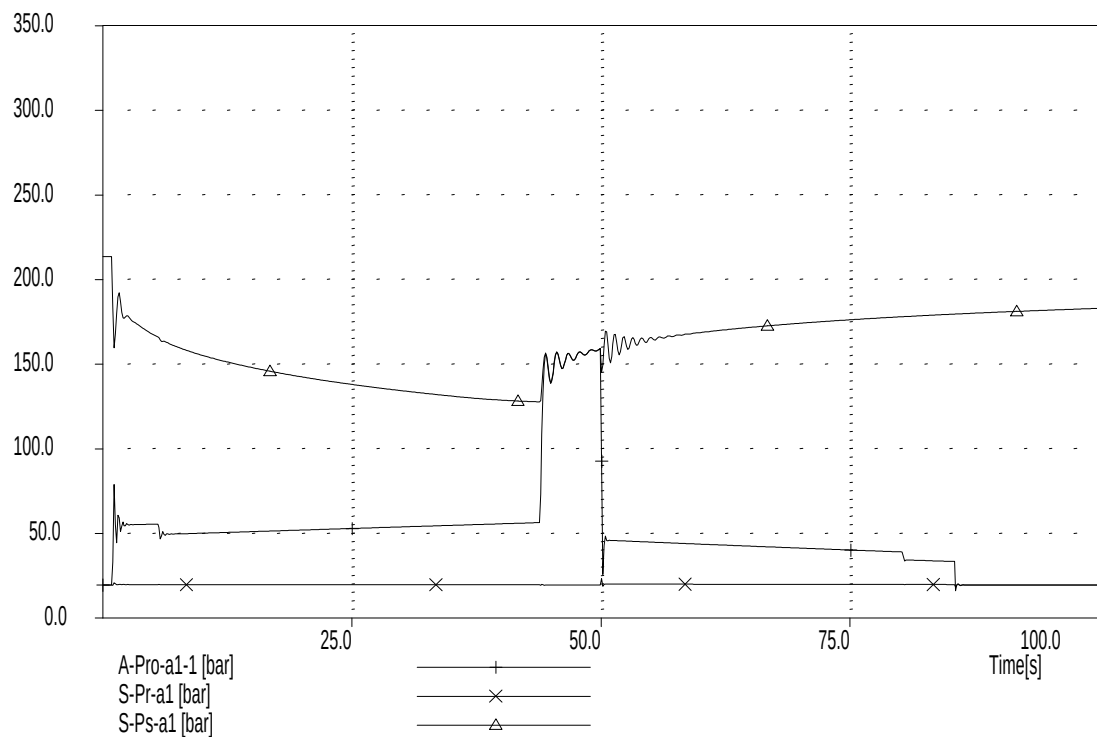


Figure 7-2 The M130 - 5" gate valve, SCM pressures

Legend: A-Pro-a1-1; Actuator opening cavity pressure (bar)
 S-Pr-a1; SCM return pressure (bar)
 S-Ps-a1; SCM supply pressure (bar)

In the plot below is shown the pressure drop across the actuator piston together with the actuator position while the valve is stroked fully open/fully closed. As can be seen, the actuator starts to close at a actuator piston pressure drop of approximately 35 bar which will be the start-to-creep pressure of the 5" valve when the bore pressure is 89 bar.

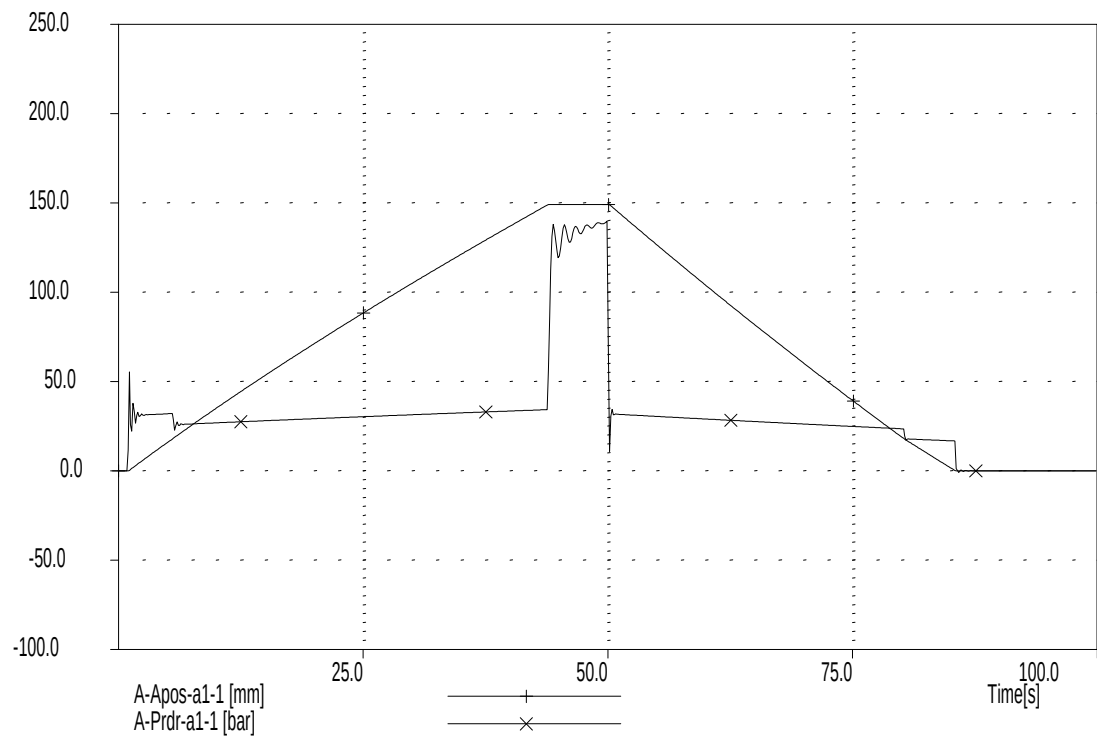


Figure 7-3 The M130 - 5" gate valve, piston pressure drop

Legend: A-Apos-a1-1; Actuator position (mm)
A-Prdr-a1-1; Actuator piston pressure drop (bar)

7.1.2 Case 1b, Operating a M130 - 2" valve

The characteristics of the M130 - 2" 690 bar gate valve is illustrated here. The first plot shows the opening and closing cavity pressures and the actuator position during operation of the valve. The open stroke takes 3.8 seconds (between $t = 1$ and 4.8 seconds) and the closing stroke takes 2 seconds (between $t = 10$ and 12 seconds). This opening and closing times is within the design requirements stated in chapter 4 of this document.

During the opening stroke the friction is at maximum before the pinch open point (gate bore connects to presses bore) is reached at $t = 1.7$ seconds. At this point the pressure across the gate is equalized and thus the friction is reduced as is reflected in a reduction in the opening cavity pressure (and increase in the return cavity pressure). Reaching the end position at $t = 4.8$ seconds, the opening cavity pressure increases to SCM actual supply pressure.

The opening cavity pressure during the closing stroke is less than during the opening stroke due to the spring force working to close the actuator. At $t = 11.5$ seconds the pinch point is reached and the friction across the gate increases as the gate pressure drop is increased. The effect of this is a drop in actuator opening cavity pressure and actuator velocity.

Observe that normal operation of valves will be with equal pressure on both sides of the gate (no pressure drop over the gate) and thus, minimized gate friction.

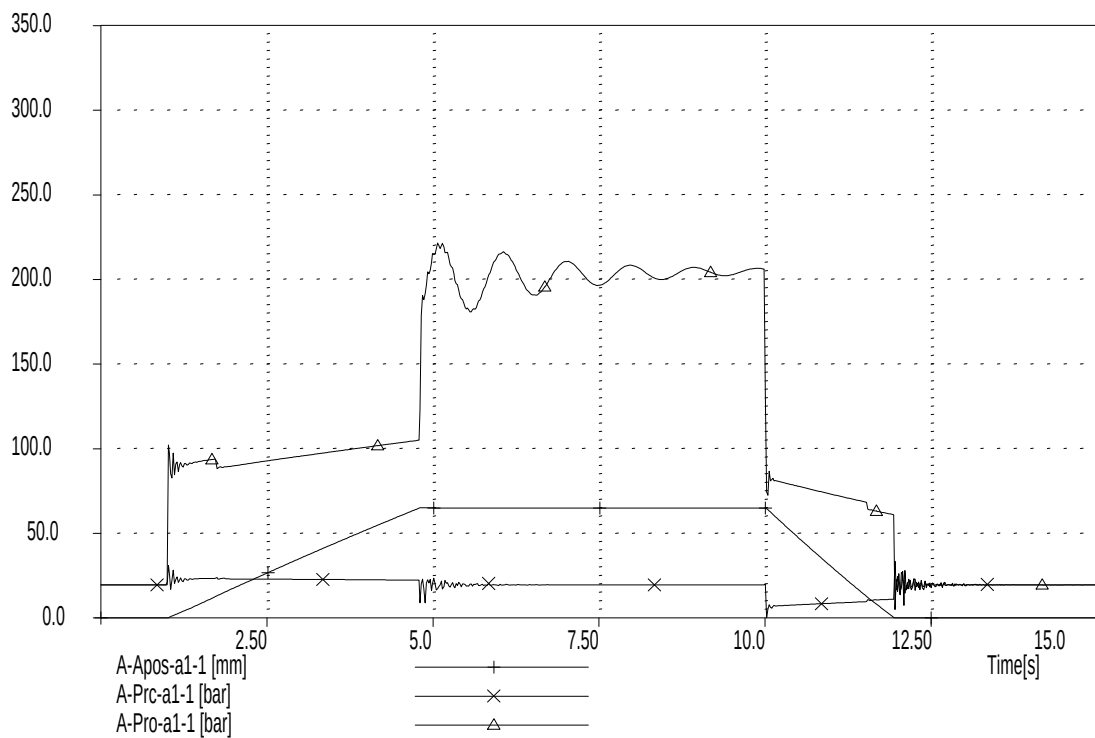


Figure 7-4 Operation of a M130 - 2" gate valve

Legend: A-Apos-a1-1; Actuator position (mm)
 A-Prc-a1-1; Actuator closing cavity pressure (bar)
 A-Pro-a1-1; Actuator opening cavity pressure (bar)

The actuator opening cavity pressure and the SCM supply and return pressures are visualized in the plot below.

The SCM supply pressure drops from 208 bar to some 180 bar during stroking the 1-liter actuator. This is well above the reset pressure of the actuator, which is approx. 35 bar according to figure 7-4.

The SCM return pressure will stabilize at approximately 20 bar during the opening of the valve and the minimum pressure difference between the SCM supply and return pressure is $180 - 20 = 160$ bar. This is above the reset pressure for the control valves which is $60 \times 1.5 = 97.5$ bar according to the design requirements.

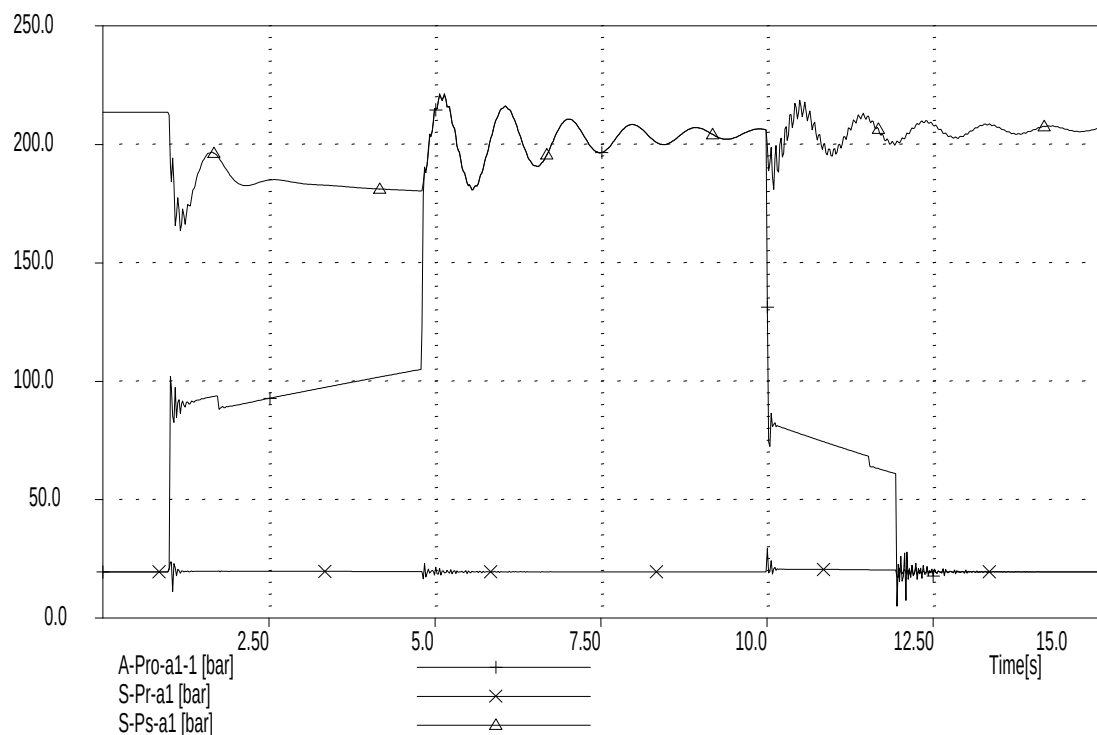


Figure 7-5 The M130 - 2" gate valve, SCM pressures

Legend: A-Pro-a1-1; Actuator opening cavity pressure (bar)
 S-Pr-a1; SCM return pressure (bar)
 S-Ps-a1; SCM supply pressure (bar)

In the plot below is shown the pressure drop across the actuator piston together with the actuator position while the valve is stroked fully open/fully closed. As can be seen, the actuator starts to close at a actuator piston pressure drop of approximately 35 bar which will be the start-to-creep pressure of the 2" valve when the bore pressure is 89 bar.

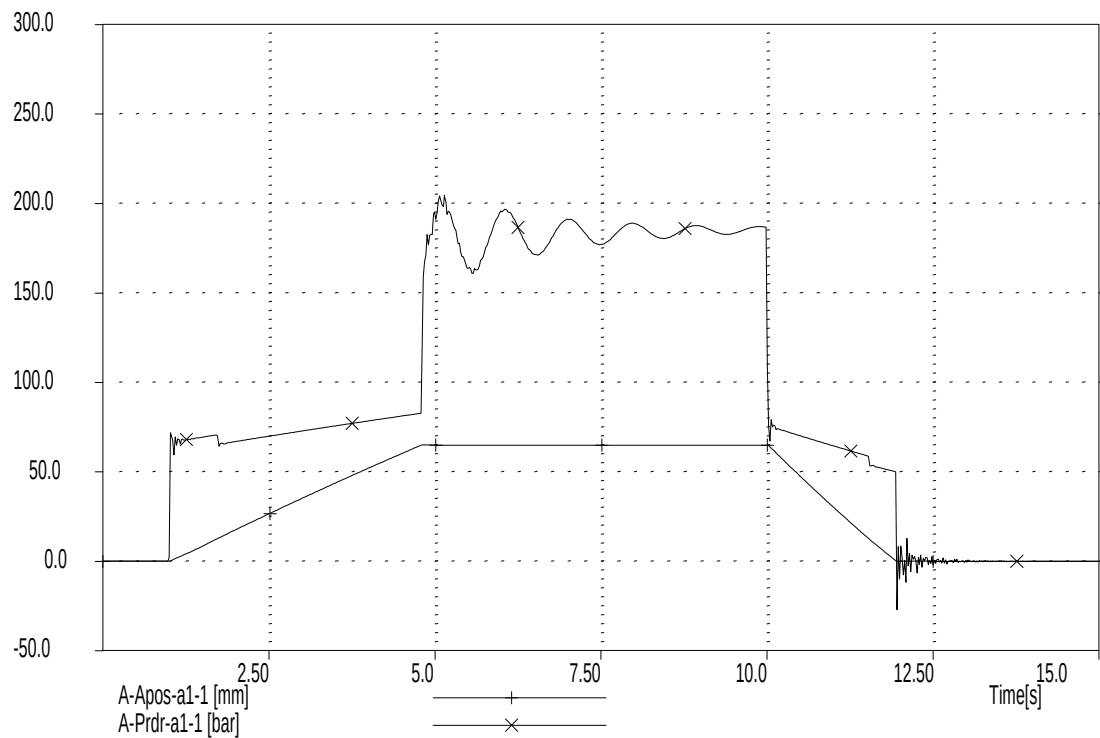


Figure 7-6 The M130 - 2" gate valve, piston pressure drop

Legend: A-Apos-a1-1; Actuator position (mm)
A-Prdr-a1-1; Actuator piston pressure drop (bar)

7.1.3 Case 1c, Operating a M140 - 1" valve

The characteristics of the M140 1" 896 bar gate valve is illustrated here. The first plot shows the opening and closing cavity pressures and the actuator position during operation of the valve. The open stroke takes 0.6 seconds (between $t = 1$ and 1.6 seconds) and the closing stroke takes 0.6 seconds (between $t = 5$ and 5.6 seconds). Due to short opening and closing time, it is not possible to see variations in pressures as results of differences in frictions and equalized pressures.

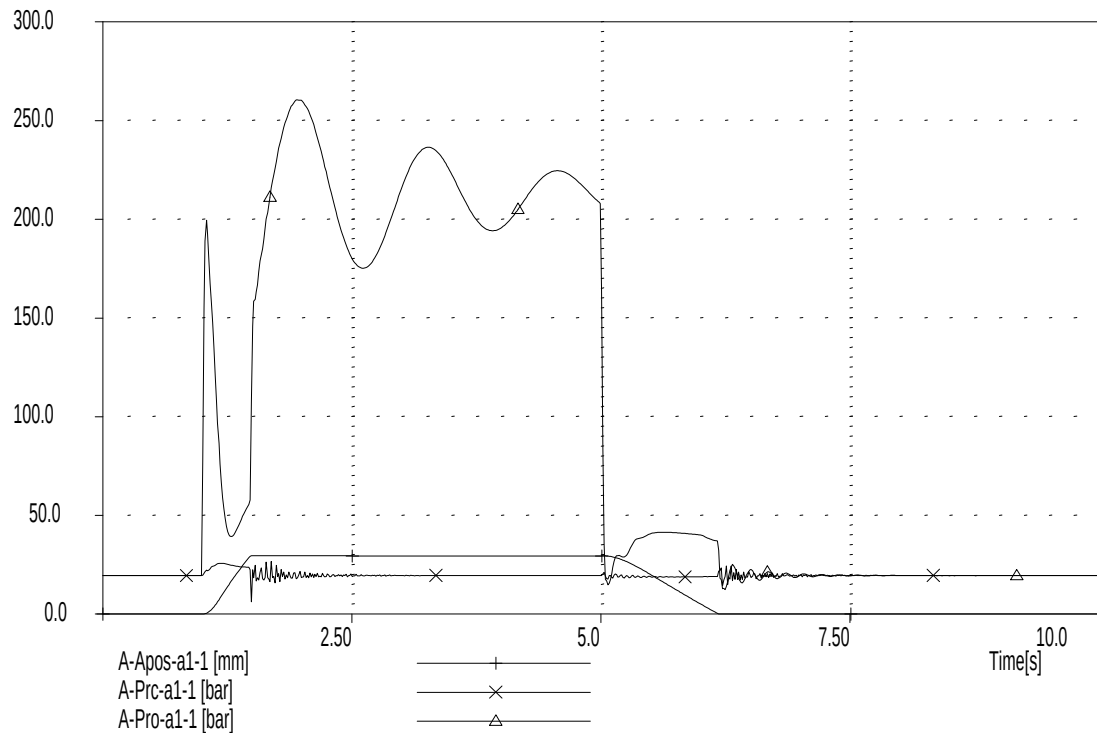


Figure 7-7 Operation of a M140 - 1" gate valve

Legend: A-Apos-a1-1; Actuator position (mm)
 A-Prc-a1-1; Actuator closing cavity pressure (bar)
 A-Pro-a1-1; Actuator opening cavity pressure (bar)

The actuator opening cavity pressure and the SCM supply and return pressures are visualized in the plot below.

The SCM supply pressure drops from 208 bar to some 180 bar during stroking the 0.5-liter actuator. This is well above the reset pressure of the actuator.

The SCM return pressure will stabilize at approximately 20 bar during the opening of the valve and the minimum pressure difference between the SCM supply and return pressure is $180 - 20 = 160$ bar. This is above the reset pressure for the control valves which is $60 \times 1.5 = 97.5$ bar according to the design requirements.

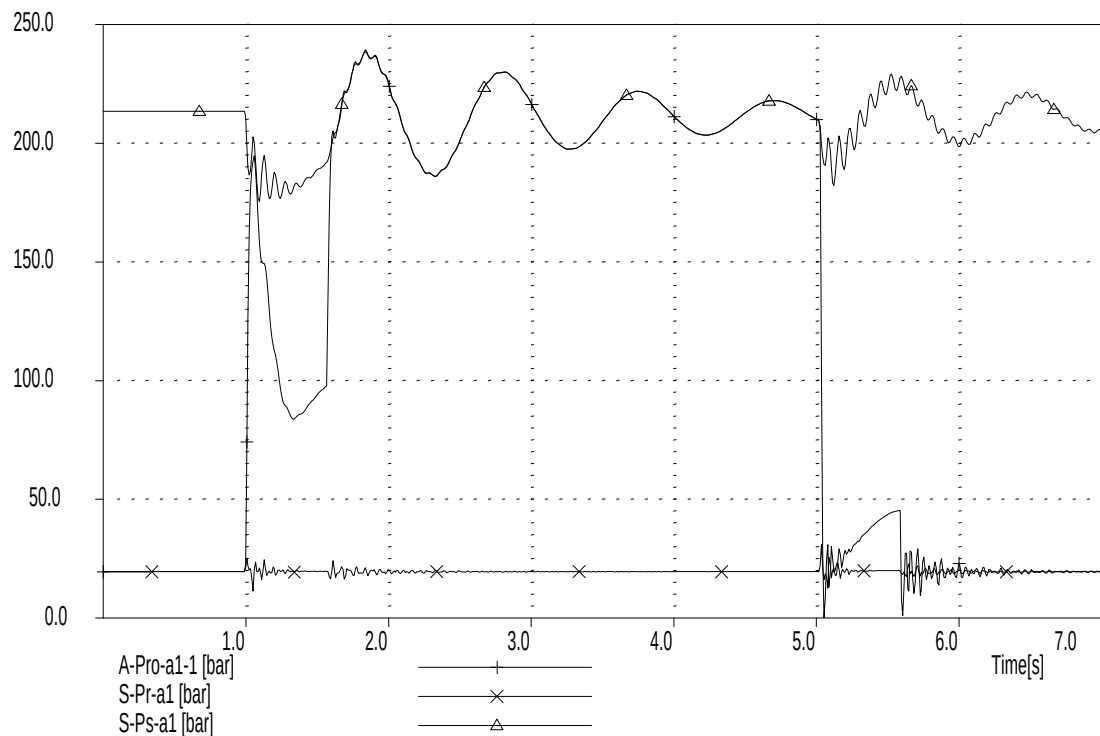


Figure 7-8 The M140 - 1" gate valve, SCM pressures

Legend: A-Pro-a1-1; Actuator opening cavity pressure (bar)
 S-Pr-a1; SCM return pressure (bar)
 S-Ps-a1; SCM supply pressure (bar)

In the plot below is shown the pressure drop across the actuator piston together with the actuator position while the valve is stroked fully open/fully closed. As can be seen, the actuator starts to close at an actuator piston pressure drop of approximately 0 bar which will be the start-to-creep pressure of the 1" valve when the bore pressure is 89 bar.

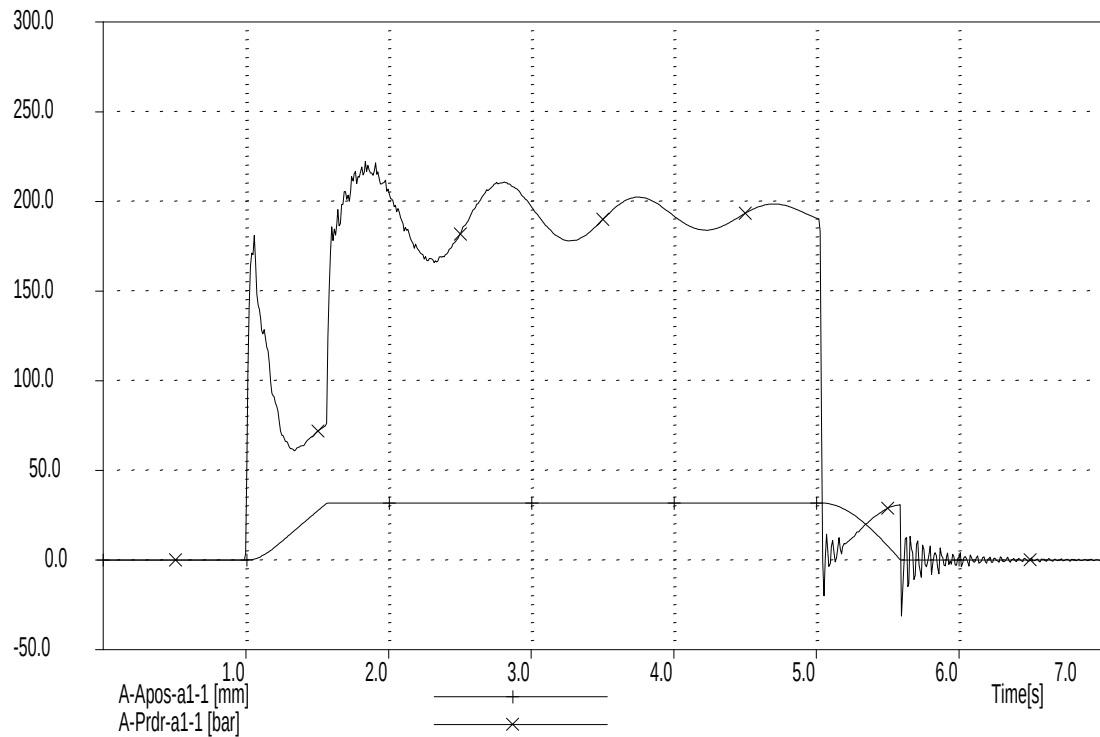


Figure 7-9 The M140 - 1" gate valve, piston pressure drop

Legend: A-Apos-a1-1; Actuator position (mm)
A-Prdr-a1-1; Actuator piston pressure drop (bar)

7.2 CASE 2, OPERATING A XT

7.2.1 Case 2a, opening a XT in sequence

During opening of two 5", two 2" and two 1" valves with a single tree on the template. The opening sequence is as follows:

- t = 1 sec. open first 5" valve
- t = 200 sec. open first 2" valve
- t = 400 sec. open last 5" valve
- t = 600 sec. open last 2" valve
- t = 700 sec. open first 1" valve
- t = 710 sec. open last 1" valve

The lowest pressure difference between supply and return lines in the SCM is estimated to approximately 105 bar as shown below. This is within acceptable limits according to the design requirements in chapter 4 of this document.

The total opening time of one XT is within acceptable limits according to the design requirements in chapter 4 of this document.

Observe that the valve operations are executed with equalized well and manifold pressures, i.e. no pressure difference across the gate of the valve.

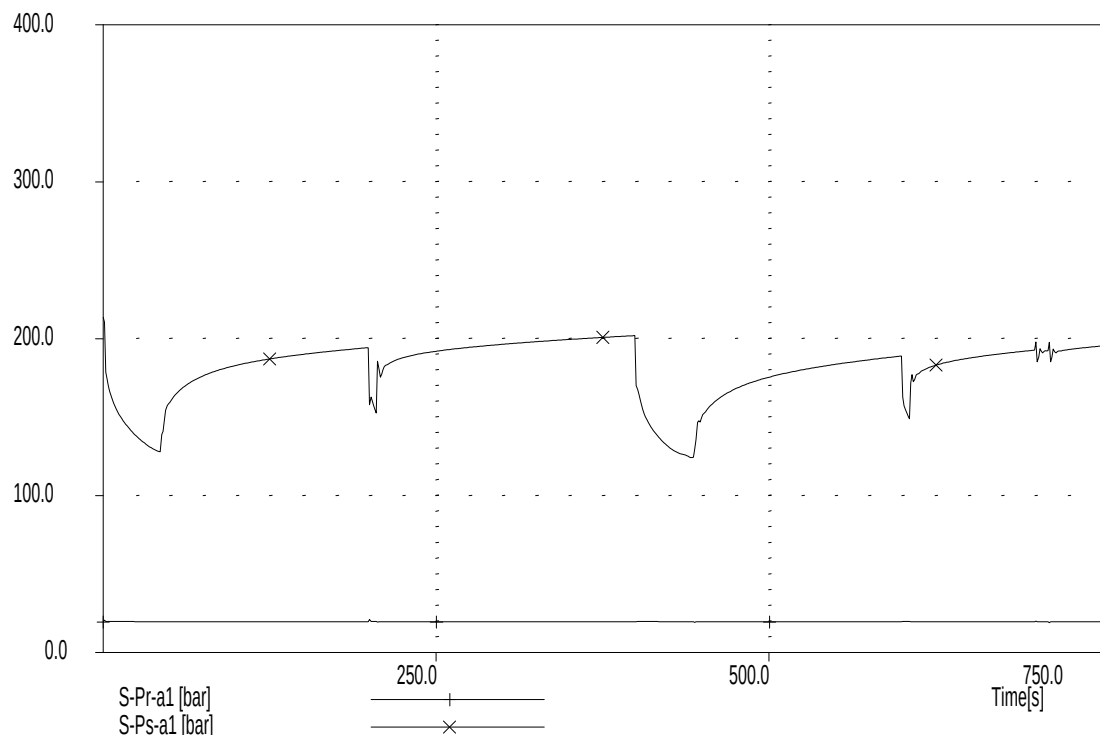


Figure 7-10 SCM Supply pressure

Legend: S-Pr-a1; SCM return pressure (bar)
 S-Ps-a1; SCM supply pressure (bar)

Below is shown the LP supply inlet (topside) and the outlet (subsea) flow rates.

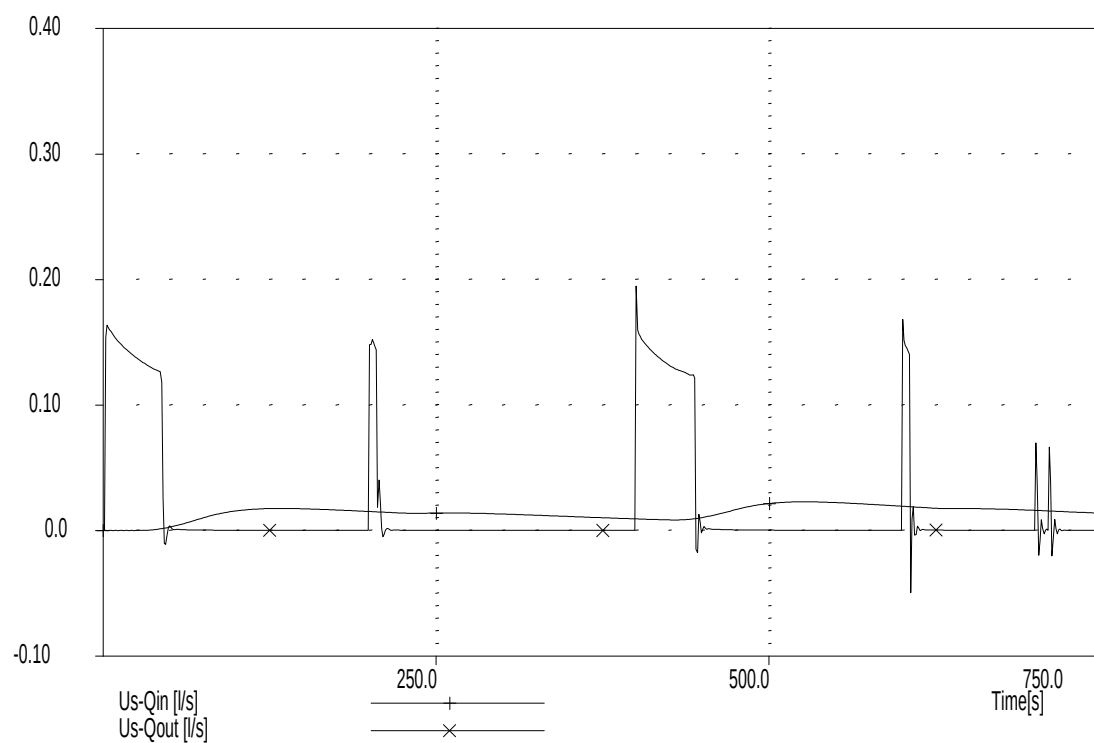


Figure 7-11 Flow rates

Legend: Us-Qin; LP supply line umbilical inlet (topside) flow rate (l/s)
Us-Qout; LP supply line umbilical outlet (subsea) flow rate (l/s)

7.2.2 Case 2b, Close down a XT controlled

The following plot shows a controlled shut down of a XT. The shut down sequence is an arbitrary chosen sequence only to show that the design requirements regarding controlled shut down is fulfilled. The closing sequence is as follows:

- t = 1 sec. close first 5" valve
- t = 50 sec. close first 2" valve
- t = 80 sec. close last 5" valve
- t = 120 sec. close last 2" valve
- t = 150 sec. close first 1" valve
- t = 160 sec. close last 1" valve

Due to limitations in the plotting software is not the first 1" valve shown at t=150 sec.

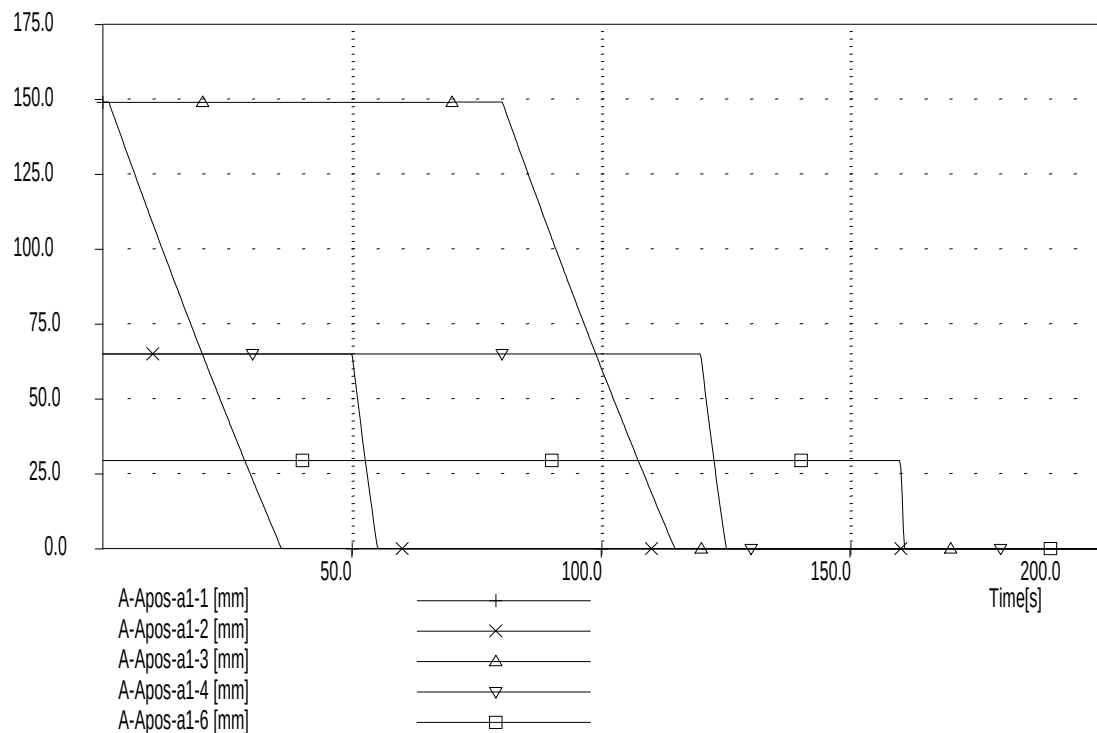


Figure 7-12 XT closing sequence

Legend: A-Apos-a1-1; Actuator position (mm)
 A-Apos-a1-2; Actuator position (mm)
 A-Apos-a1-3; Actuator position (mm)
 A-Apos-a1-4; Actuator position (mm)
 A-Apos-a1-6; Actuator position (mm)

7.3 CASE 3, CHOKE OPERATION

The plot below shows the SCM supply pressure while operating one choke. The choke is a Masterflo-P4 balanced choke operated at 3.5 seconds per step (1.5 seconds to open choke actuator and 2 seconds to close it).

As can be seen the minimum difference between supply and return pressure is approximately 115 bar. This is above the reset pressure for the control valves which is $60 \times 1.5 = 97.5$ bar according to the design requirements.

Choke operation speed: 3.5 seconds per step. Operate from fully opened to fully closed will take 700 seconds (12 minutes, 200 steps)

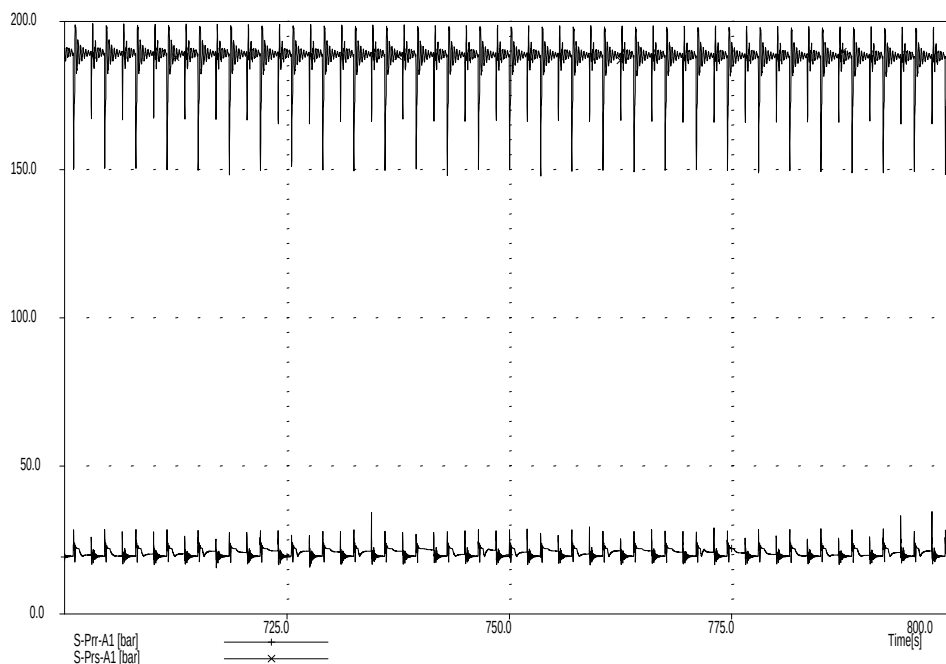


Figure 7-13 SCM supply and return pressure while operating a choke

Legend: S-Pr-a1; SCM return pressure (bar)
S-Ps-a1; SCM supply pressure (bar)

7.4 CASE 4, LP-UMBILICAL SIMULATIONS

7.4.1 Umbilical charge up

This case simulates charge-up of the supply line. The compression of fluid and expansion of hoses dominates the first part of the charge-up sequence.

At $t = 2000$ seconds will the SCM supply pressure be 226 bar which is 207 bar supply pressure + added pressure due to height difference. This is within the design requirements as specified in chapter 4 of this document.

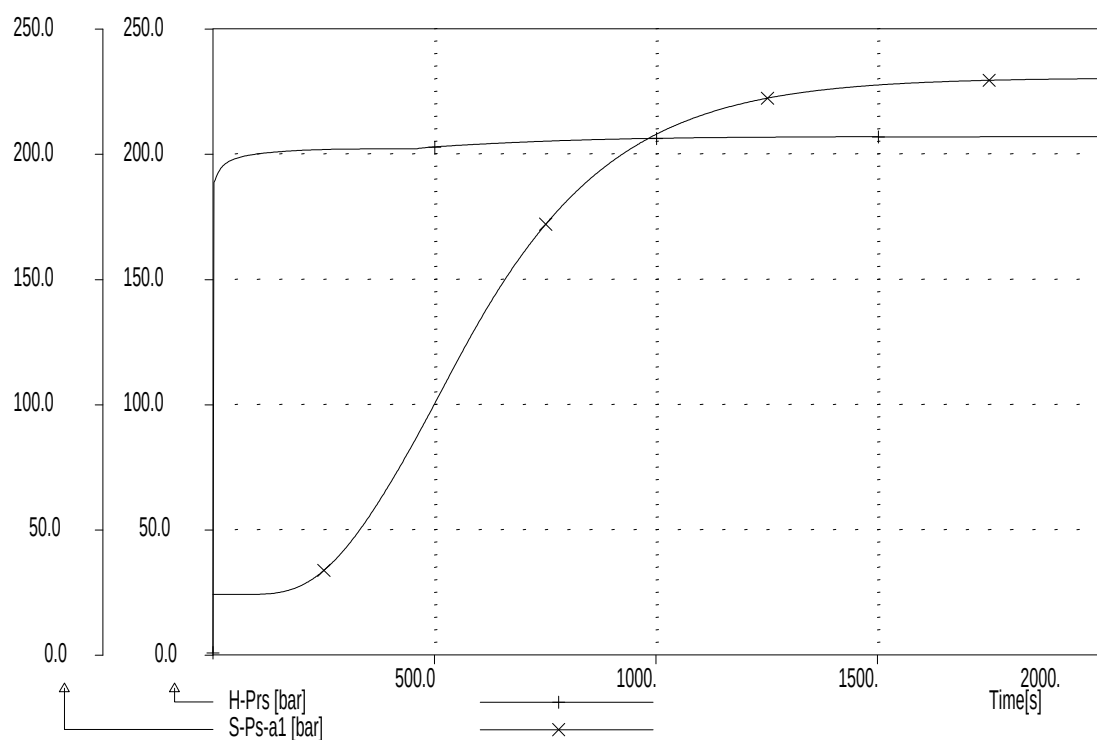


Figure 7-14 LP-umbilical charge-up, pressure response

Legend: H-Prs; HPU outlet pressure (bar)
 S-Ps-a1; SCM supply pressure (bar)

The umbilical flow rate during the charge-up is shown below together with variations in HPU accumulator gas volume. The HPU accumulator precharge volume is 220 liter. As can be seen the HPU accumulators help charging-up the umbilical together with the pumps.

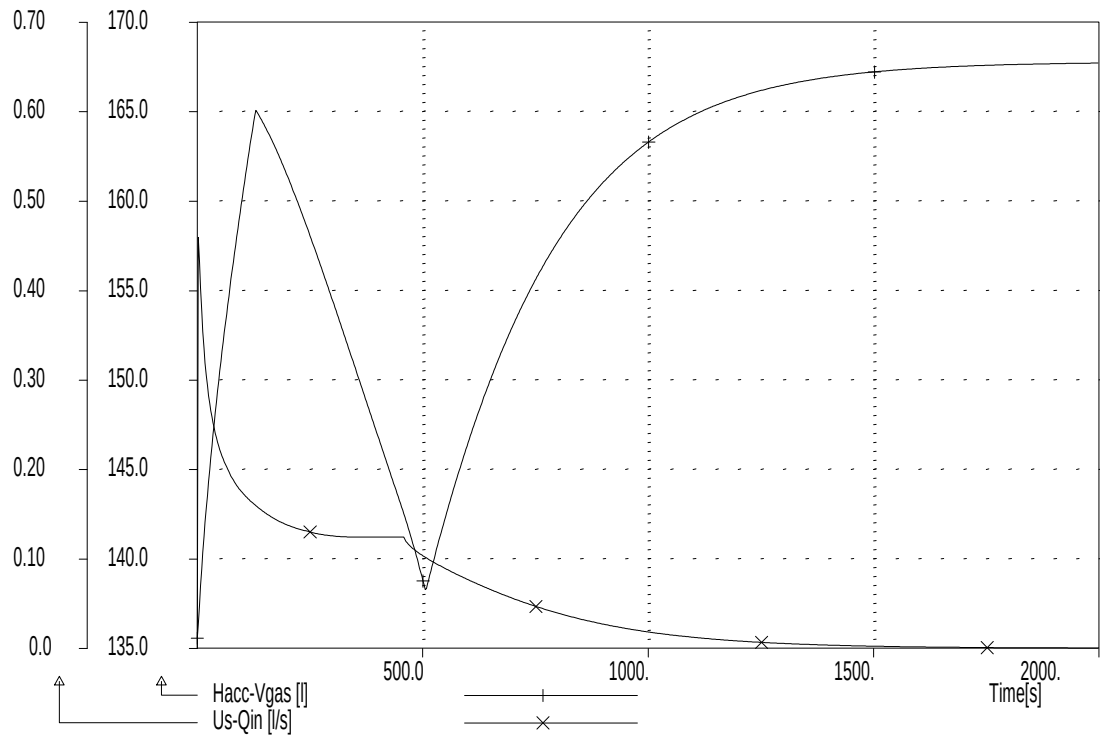


Figure 7-15 LP umbilical charge-up, flow rates

Legend: Hacc-Vgas; Gas volume in HPU accumulators (liter)
Us-Qin; Umbilical inlet flow rate (l/s)

7.4.2 Umbilical bleed down (ESD)

This case simulates bleed-down of the supply line with 2 off 5", 2 off 2" and 2 off 1" valves on each of the 5 XT's. The compressed fluid being drained dominates the first part of a bleed-down sequence. Once the pressure is down to the cracking pressure of the actuators, the SCM supply pressure increase slightly as the actuators start to move.

The plot below shows the SCM supply pressure during this operation. At $t = 1000$ seconds will the supply pressure be 55 bar and the control valves will be reset at approx. 65 bar. I.e. the ESD sequence will last approx. 1000 sec.

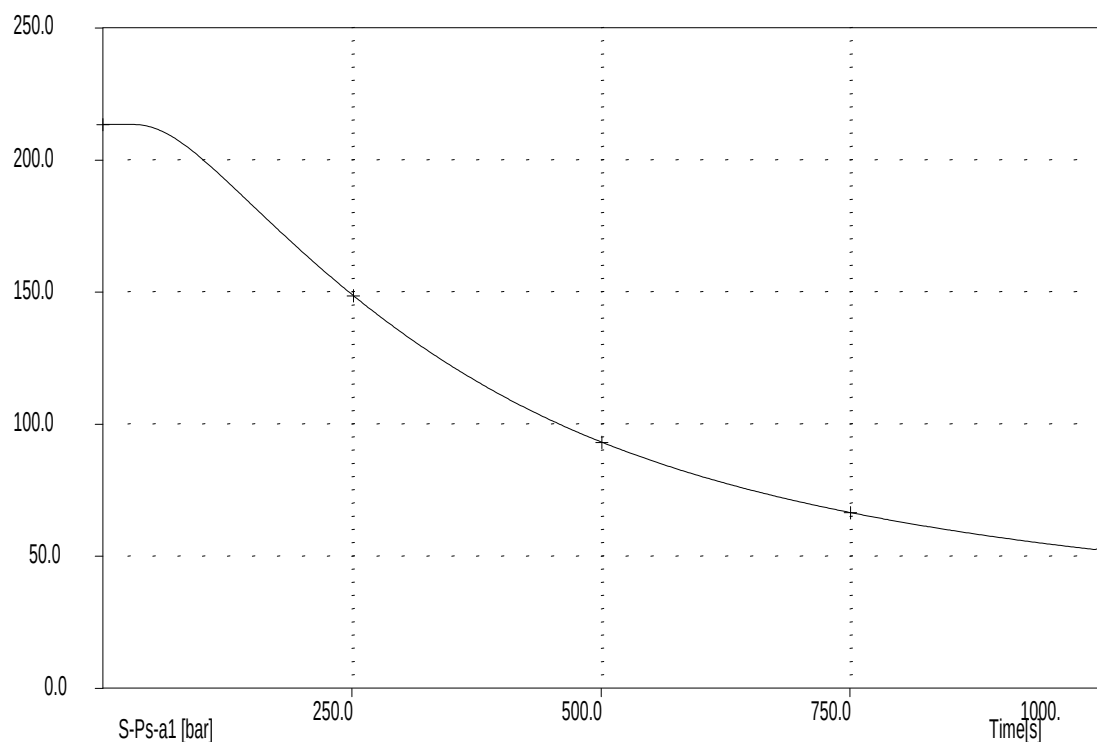


Figure 7-16 SCM supply pressure during bleed-down

Legend: S-Ps-a1; SCM supply pressure (bar)

The umbilical flow rate during the bleed-down is shown below. As can be seen the peak flow out of the umbilical topside is approximately 0.55 l/s. After that the flow diminishes to approximately 0.015 l/s.

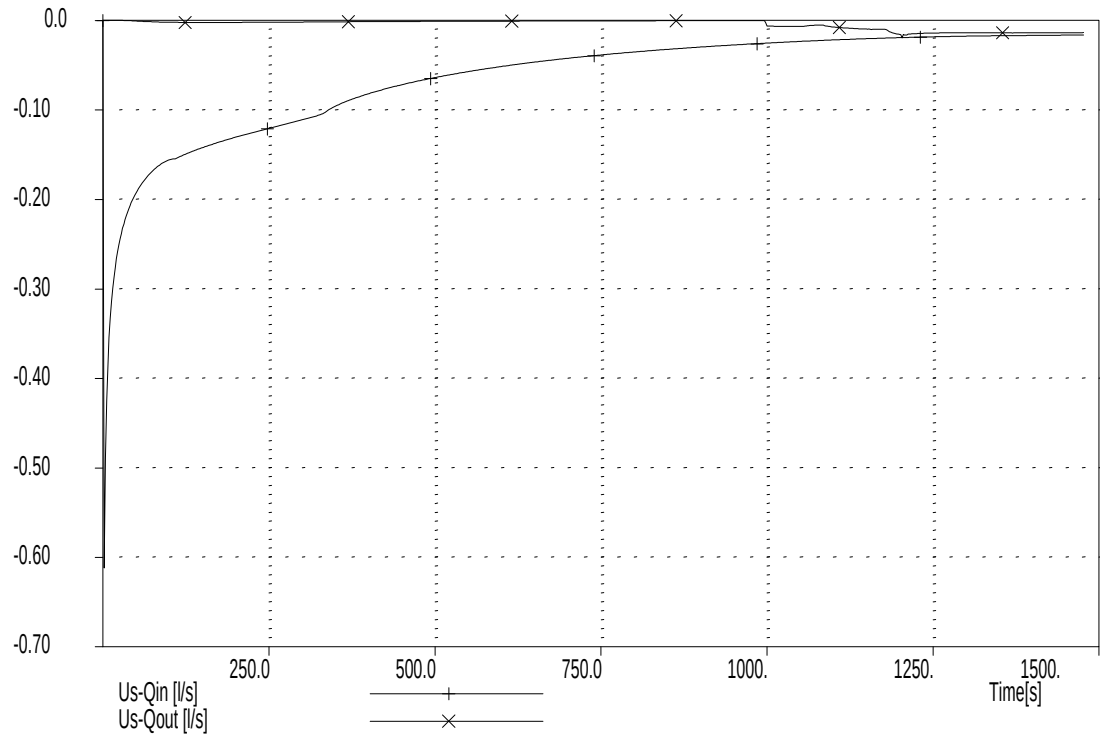


Figure 7-17 LP umbilical bleed-down, flow rates

Legend: Us-Qin; Umbilical outlet (subsea) flow rate (l/s)
Us-Qout; Umbilical inlet (topside) flow rate (l/s)

7.5 CASE 5, OPERATING THE SCSSV

The characteristic of the SCSSV (Baker 5") valve is illustrated in the plot below where is shown the actuator opening cavity pressure when driven first to open, and then to closed position. The well pressure is 89 bar while the manifold pressure before pinch point is reached is 1 bar.

The SCSSV model is set up with the SCM at a water depth of 184 meter and a 1/8" tube down to the SCSSV which is set 115 meter below the SCM.

The valve takes approximately 1 second to open and 1.2 seconds to close.

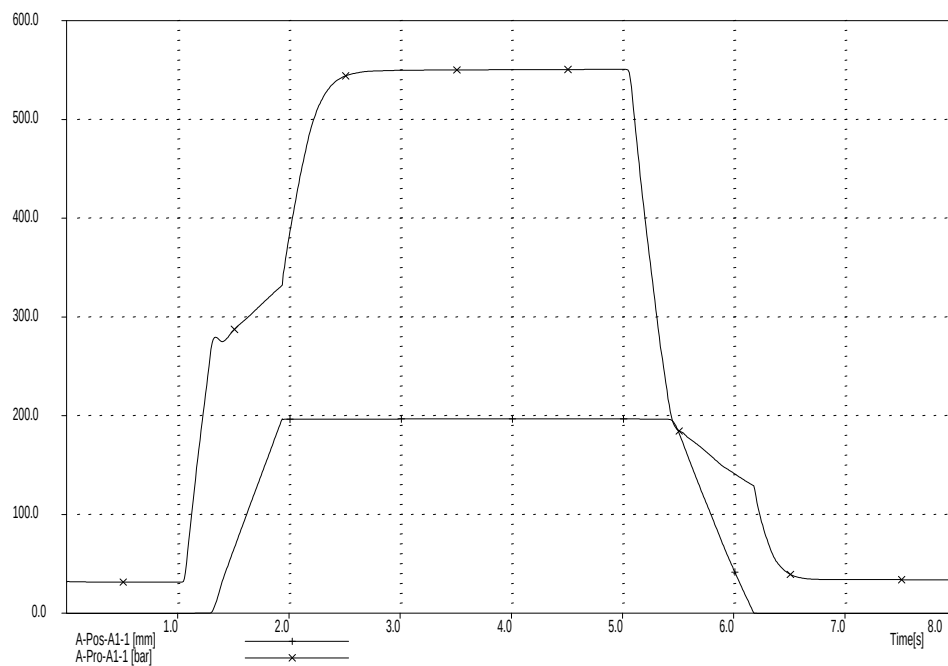


Figure 7-18 Operating the SCSSV

Legend: A-Apos-a1-1; SCSSV actuator position (mm)
A-Pro-a1-1; SCSSV open cavity pressure (bar)

7.6 CASE 6, HP-UMBILICAL SIMULATIONS

7.6.1 Umbilical charge-up

This case simulates charge-up of the supply line. The case is executed with ½” and 3/8” umbilical ID. The two cases are presented in the same plots to visualize the difference.

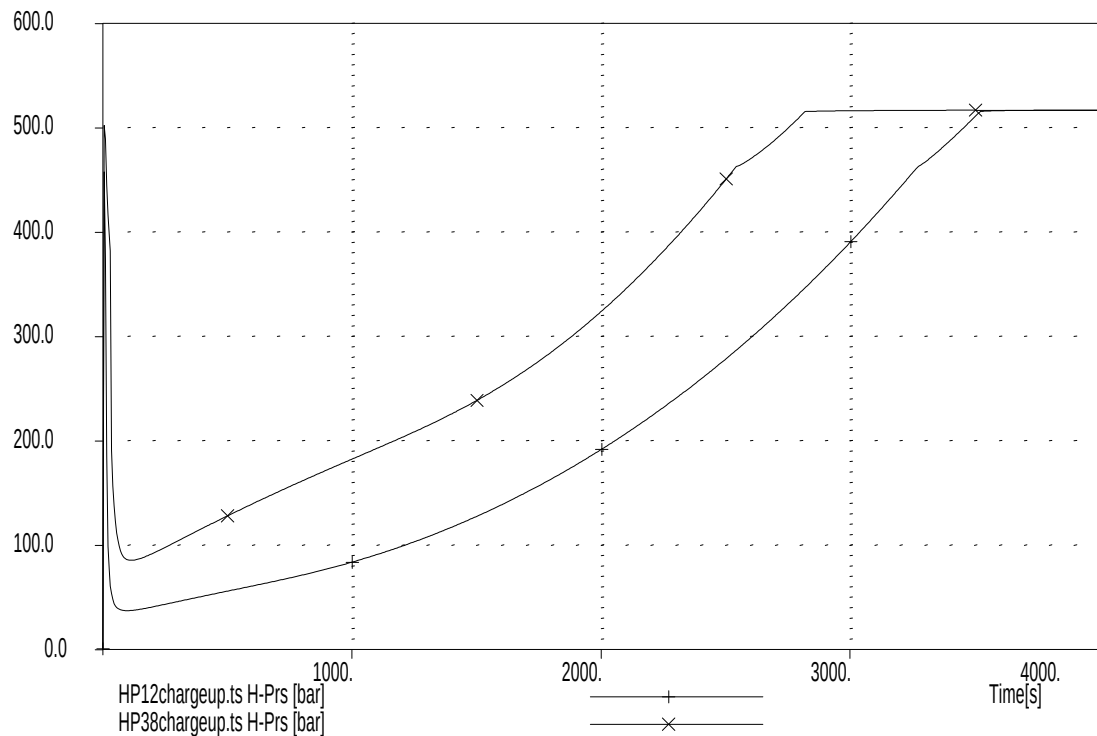


Figure 7-19 HP-umbilical charge-up, topside pressure response

Legend: HP12chargeup.ts H-Prs; HPU outlet pressure – ½” ID in umbilical (bar)
 HP38chargeup.ts H-Prs; HPU outlet pressure – 3/8” ID in umbilical (bar)

The subsea pressure response shows that the design requirements stated in chapter 4 of this document not will be met with either ½” or 3/8” umbilical ID. It will therefor be recommended to increase the HPU accumulator or pump capacity.

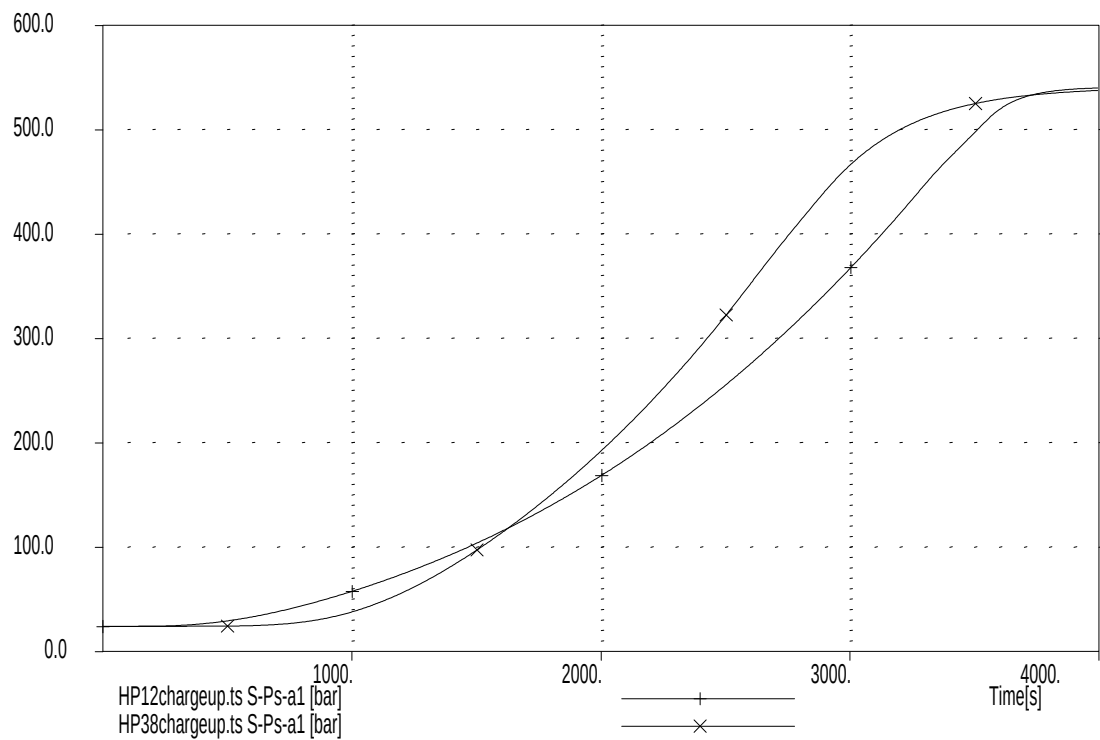


Figure 7-20 HP-umbilical charge-up, subsea pressure response

Legend: HP12chargeup.ts H-Prs; HPU outlet pressure – ½" ID in umbilical (bar)
HP38chargeup.ts H-Prs; HPU outlet pressure – 3/8" ID in umbilical (bar)

The umbilical flow rate during the charge-up of the 3/8" umbilical ID is shown below together with variations in HPU accumulator gas volume. The HPU accumulator precharge volume is 44 liter. As can be seen the HPU gas volume raises fast up to 44 liter which means that the accumulator is empty. The charge up time is then depending on the pump capacity.

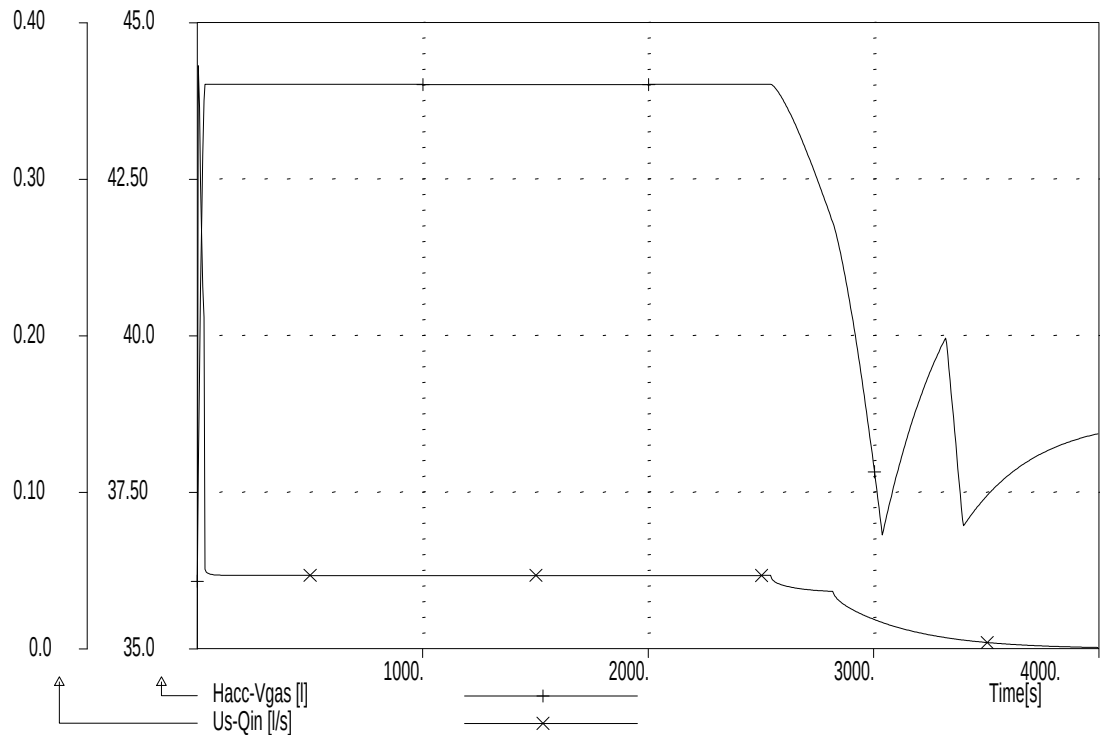


Figure 7-21 HP umbilical charge-up, flow rates

Legend: Hacc-Vgas; Gas volume in HPU accumulators (liter)
 Us-Qin; Umbilical inlet flow rate (l/s)

7.6.2 HP Umbilical charge up – With recommended modifications

This case simulates charge-up of the HP supply line with recommended modifications which is reduced umbilical ID from ½” to 3/8” and increased the HPU pump capacity from 2.8 l/min to 4.2 l/min.

The compression of fluid and expansion of hoses dominates the first part of the charge-up sequence. At $t = 3000$ seconds will the SCM supply pressure be 536 bar which is 517 bar supply pressure + added pressure due to height difference. This is within the design requirements as specified in chapter 4 of this document.

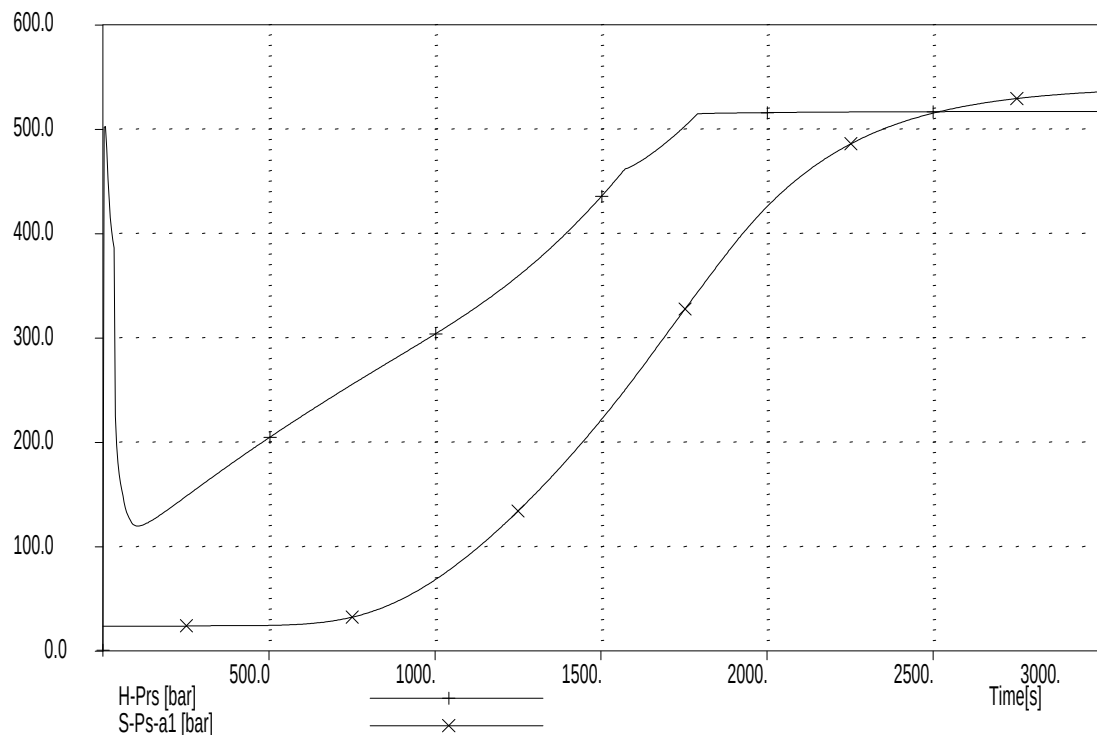


Figure 7-22 HP-umbilical charge-up, pressure response

Legend: H-Prs; HPU outlet pressure (bar)
S-Ps-a1; SCM supply pressure (bar)

The umbilical flow rate during the charge-up is shown below together with variations in HPU accumulator gas volume. The HPU accumulator precharge volume is 44 liter. As can be seen the HPU accumulators help charging-up the umbilical together with the pumps.

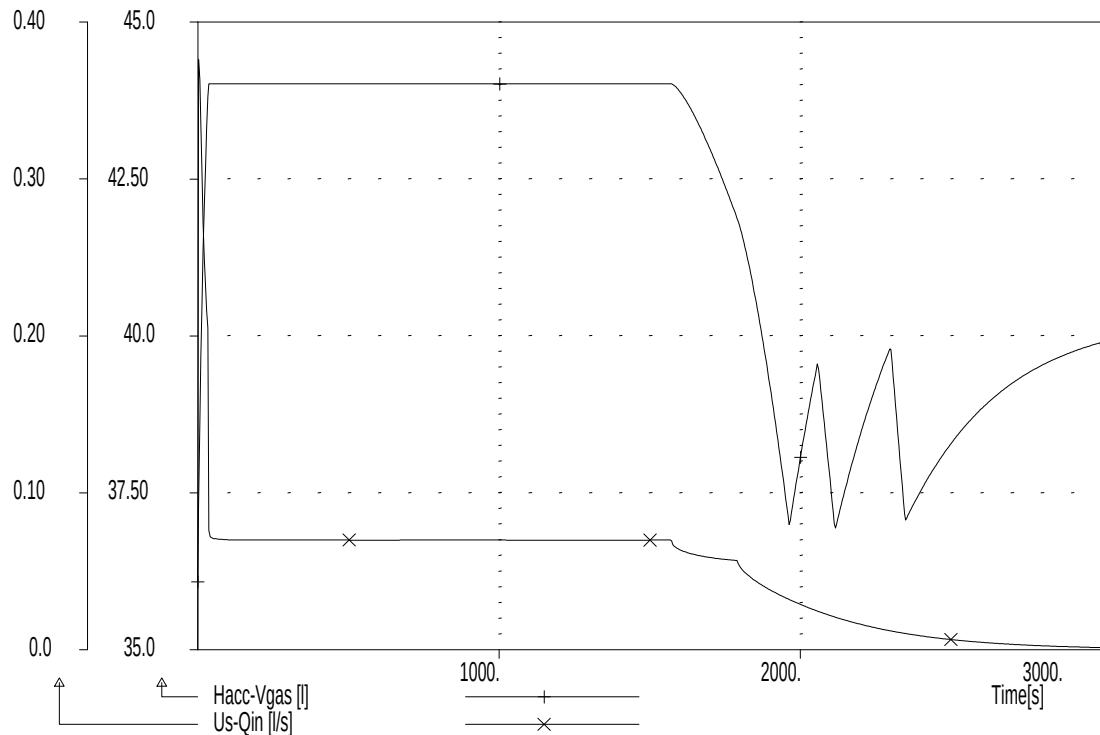


Figure 7-23 HP umbilical charge-up, flow rates

Legend: Hacc-Vgas; Gas volume in HPU accumulators (liter)
Us-Qin; Umbilical inlet flow rate (l/s)

7.6.3 Umbilical bleed-down (ESD)

This case simulates bleed-down of the ½” umbilical ID supply line, worst case, with 1 off 5” Baker SCSSV on each of the 6 XT's. The compressed fluid being drained dominates the first part of a bleed-down sequence. Once the pressure is down to the cracking pressure of the actuators, the SCM supply pressure increase slightly as the actuators start to move.

The plot below shows the SCM supply pressure during this operation together with one of the SCSSV's position. The SCSSV's are fully closed after 620 seconds.

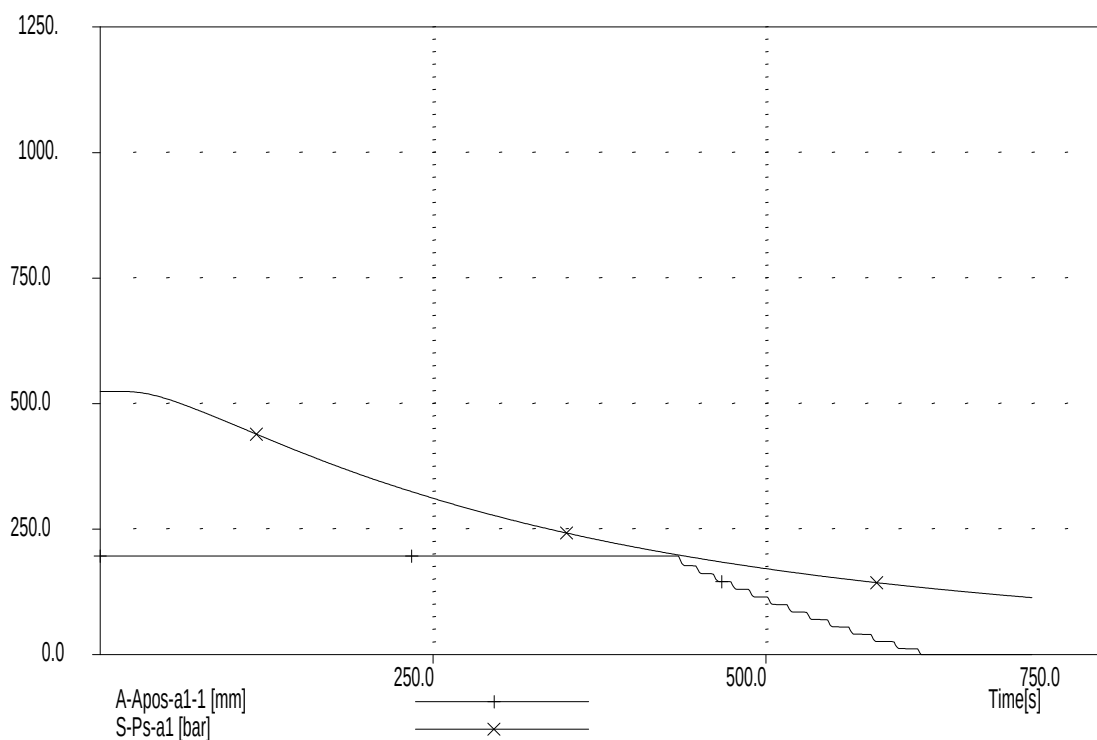


Figure 7-24 Actuator positions and SCM supply pressure during bleed-down

Legend: A-Apos-a1-1; SCSSV actuator position (mm)
 S-Ps-a1; SCM supply pressure (bar)

The umbilical flow rate during the bleed-down is shown below. As can be seen the peak flow out of the umbilical topside is approximately 0.75 l/s. After that the flow diminishes to approximately 0.015 l/s.

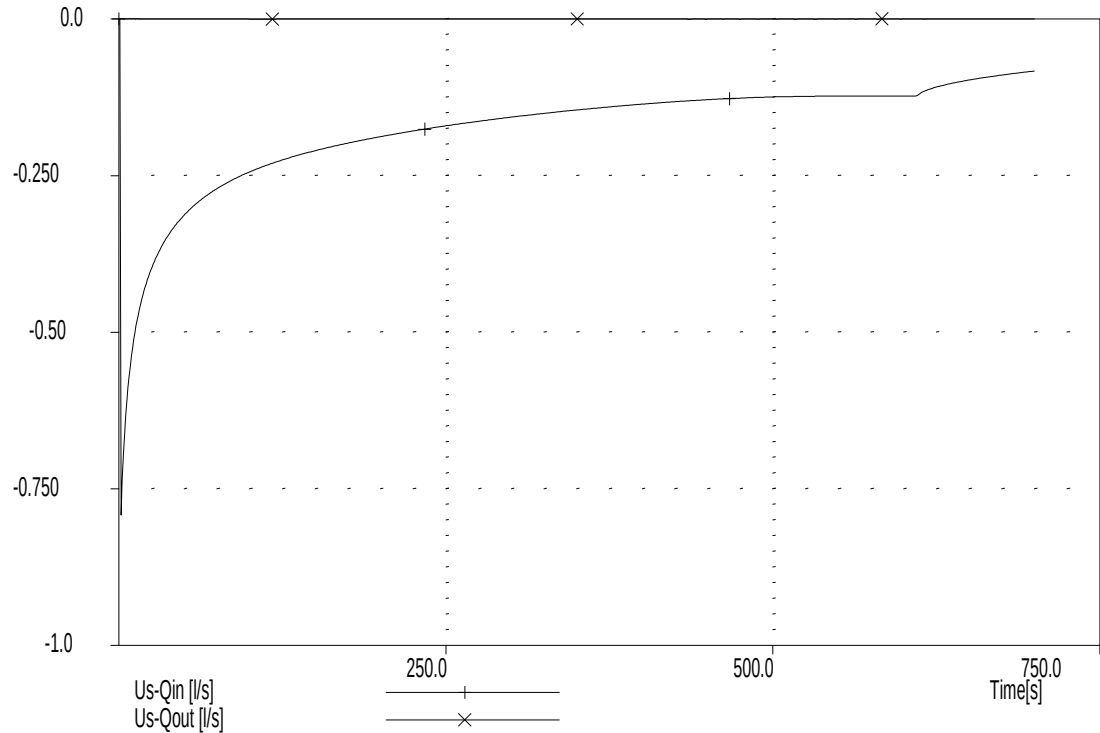


Figure 7-25 HP umbilical bleed-down, flow rates

Legend: Us-Qin; Umbilical outlet (subsea) flow rate (l/s)
Us-Qout; Umbilical inlet (topside) flow rate (l/s)

7.7 CASE 7, CHOSEN CASES WITH SUBSEA ACCUMULATOR

The following are chosen cases where a 20 liter subsea accumulator is included on the manifold.

7.7.1 Case 7a, Operating a M130 - 5" valve

The characteristics of the M130 - 5" 690 bar gate valve is illustrated here. The first plot shows the opening and closing cavity pressures and the actuator position during operation of the valve. The open stroke takes 39 seconds (between $t = 1$ and 40 seconds) and the closing stroke takes 36 seconds (between $t = 50$ and 86 seconds). The opening time is 4 seconds shorter compared to the case without subsea accumulator. The closing time is the same.

Observe that normal operation of valves will be with equal pressure on both sides of the gate (no pressure drop over the gate) and thus, minimized gate friction.

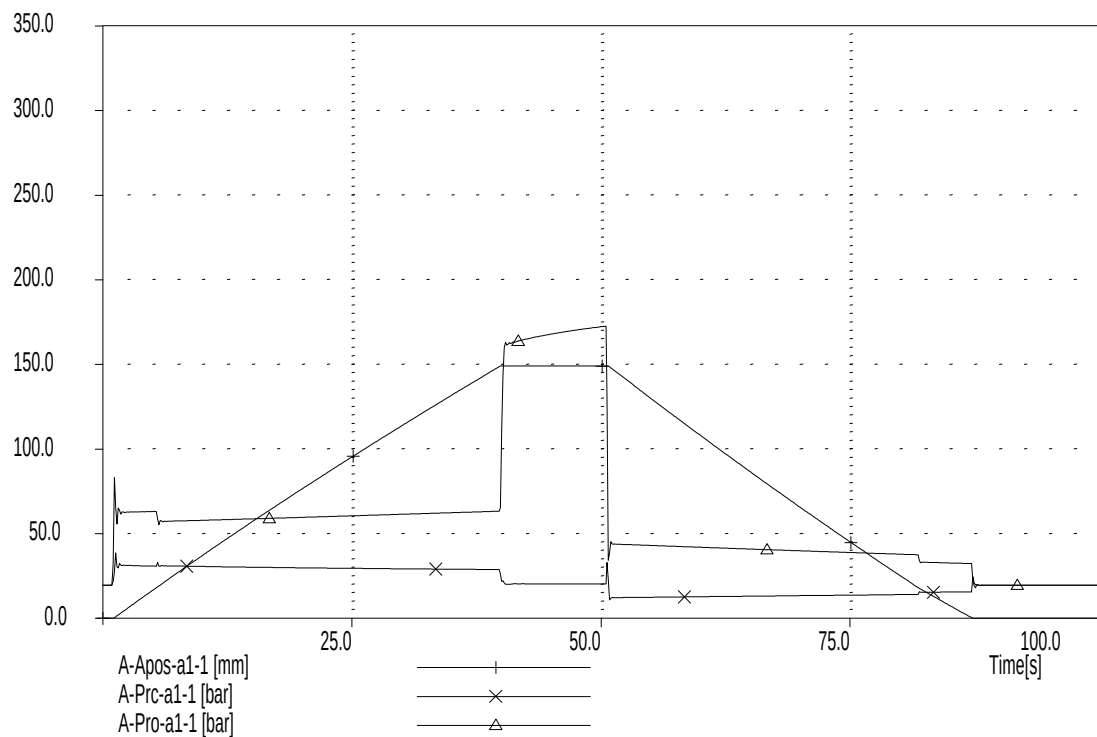


Figure 7-12 Operation of a M130 - 5" gate valve

Legend: A-Apos-a1-1; Actuator position (mm)
 A-Prc-a1-1; Actuator closing cavity pressure (bar)
 A-Pro-a1-1; Actuator opening cavity pressure (bar)

The actuator opening cavity pressure and the SCM supply and return pressures are visualized in the plot below.

The SCM supply pressure drops from 208 bar to some 155 bar during stroking the 8-liter actuator. This is 28 bar above the case without subsea accumulator (case 1a).

The SCM return pressure will stabilize at approx. 21 bar during the opening of the valve and the minimum pressure difference between the SCM supply and return pressure is $155 - 21 = 134$ bar, i.e. 28 bar better margin than case 1a.

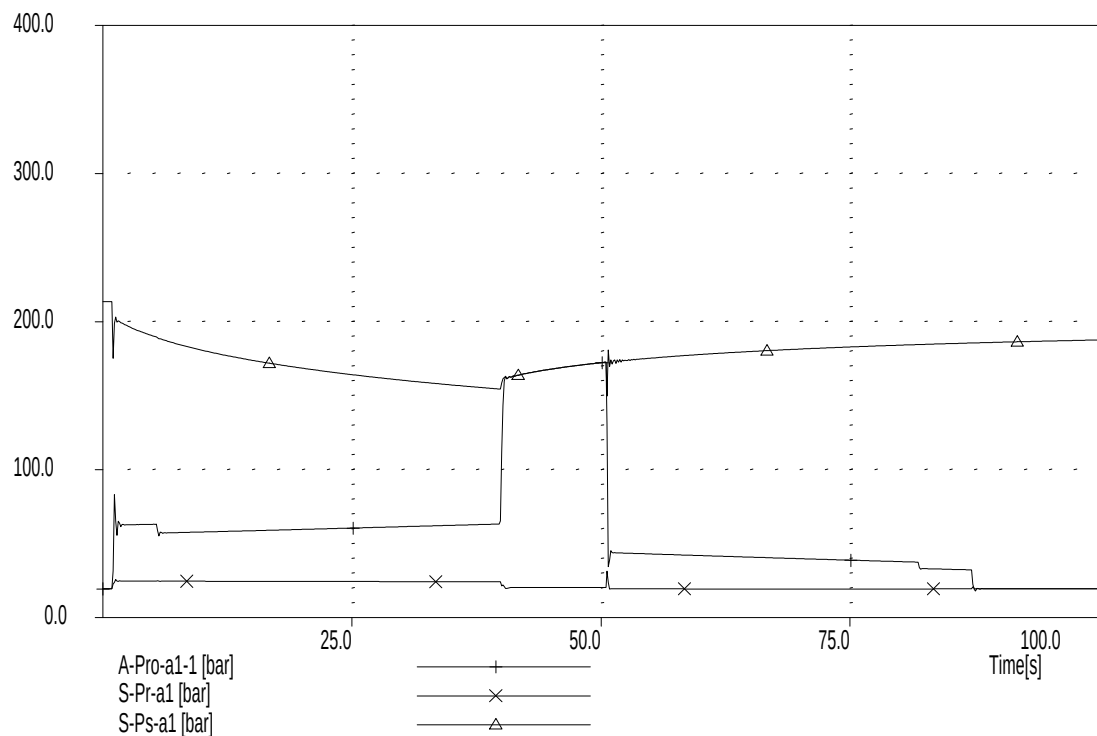


Figure 7-13 The M130 - 5" gate valve, SCM pressures

Legend: A-Pro-a1-1; Actuator opening cavity pressure (bar)
 S-Pr-a1; SCM return pressure (bar)
 S-Ps-a1; SCM supply pressure (bar)

7.7.2 Case 7b, opening a XT in sequence

During opening of two 5", two 2" and two 1" valves with a single tree on the template. The opening sequence is as follows:

- t = 1 sec. open first 5" valve
- t = 200 sec. open first 2" valve
- t = 400 sec. open last 5" valve
- t = 600 sec. open last 2" valve
- t = 700 sec. open first 1" valve
- t = 710 sec. open last 1" valve

The lowest pressure difference between supply and return lines in the SCM is estimated to approximately 124 bar as shown below. This is 19 bar better than compared to the case without subsea accumulator (case 2a).

Observe that the valve operations are executed with equalized well and manifold pressures, i.e. no pressure difference across the gate of the valve.

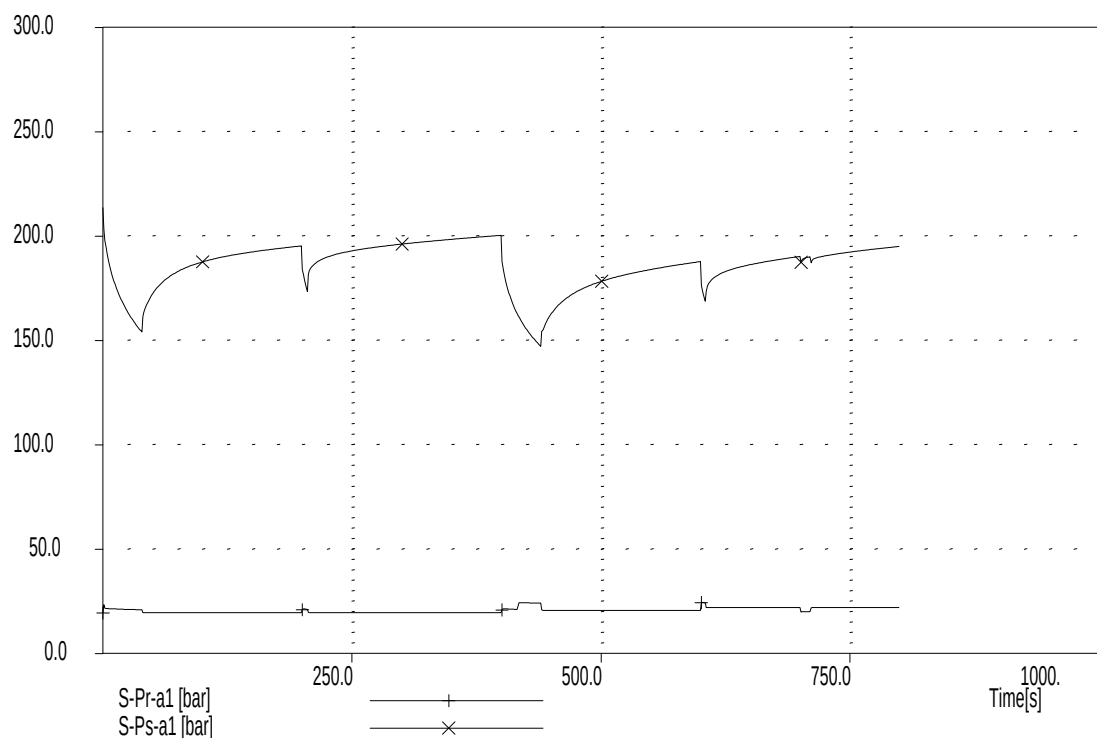


Figure 7-14 SCM Supply pressure

Legend: S-Pr-a1; SCM return pressure (bar)
 S-Ps-a1; SCM supply pressure (bar)

7.7.3 Case 7c, Choke operation

The plot below shows the SCM supply pressure while operating one choke. The choke is a Masterflo-P4 balanced choke operated at 3.5 seconds per step (1.5 seconds to open choke actuator and 2 seconds to close it).

As can be seen the minimum difference between supply and return pressure is approximately 130 bar. This is 30 bar higher than without subsea accumulator (case 3).

Choke operation speed: 3.5 seconds per step. Operate from fully opened to fully closed will take 700 seconds (12 minutes, 200 steps)

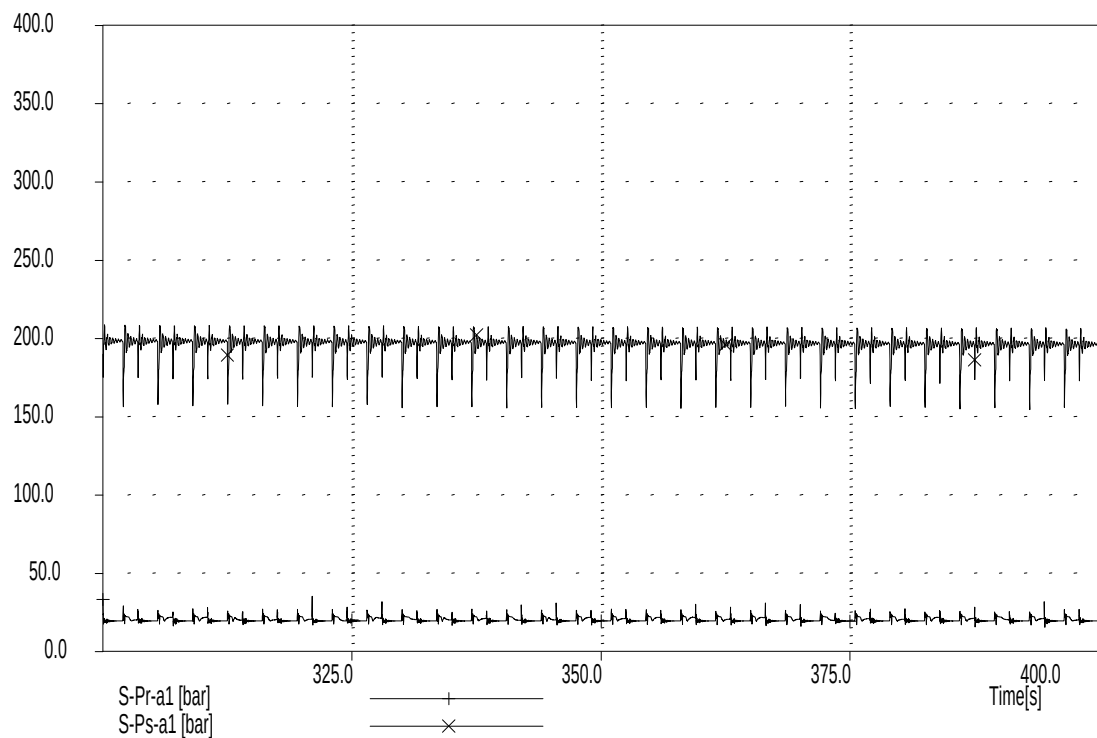


Figure 7-13 SCM supply and return pressure while operating a choke

Legend: S-Pr-a1; SCM return pressure (bar)
S-Ps-a1; SCM supply pressure (bar)

7.7.4 Case 7d, Umbilical charge up

This case simulates charge-up of the supply line. The compression of fluid and expansion of hoses dominates the first part of the charge-up sequence.

At $t = 2000$ seconds the SCM supply pressure is 213 bar which is 14 bar lower than without the subsea accumulator case(4a).

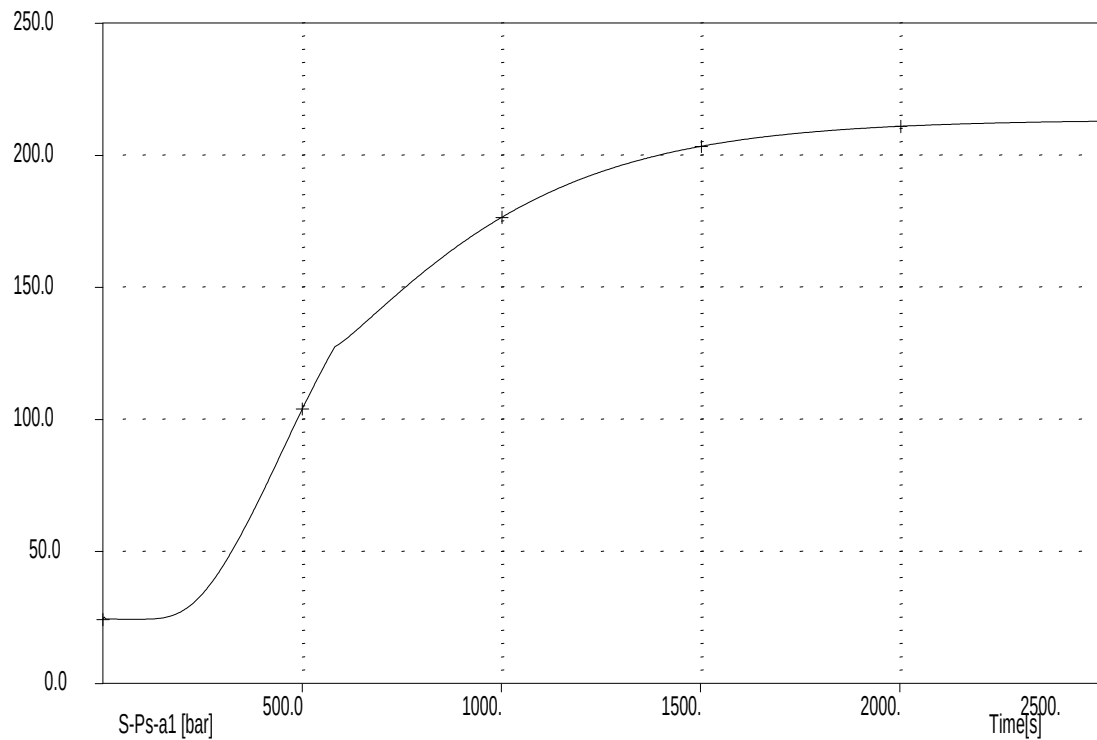


Figure 7-14 LP-umbilical charge-up, pressure response

Legend: S-Ps-a1; SCM supply pressure (bar)

7.7.5 Case 7e, Umbilical bleed down (ESD)

This case simulates bleed-down of the supply line with 2 off 5", 2 off 2" and 2 off 1" valves on each of the 5 XTs. The compressed fluid being drained dominates the first part of a bleed-down sequence. Once the pressure is down to the cracking pressure of the actuators, the SCM supply pressure increase slightly as the actuators start to move.

The plot below shows the SCM supply pressure during this operation. At $t = 1000$ seconds the supply pressure will be 55 bar. This is the same result as in the case without subsea accumulator (case 4b). The reason for this is that the 20 liter subsea accumulator represents a small amount of fluid in the total system.

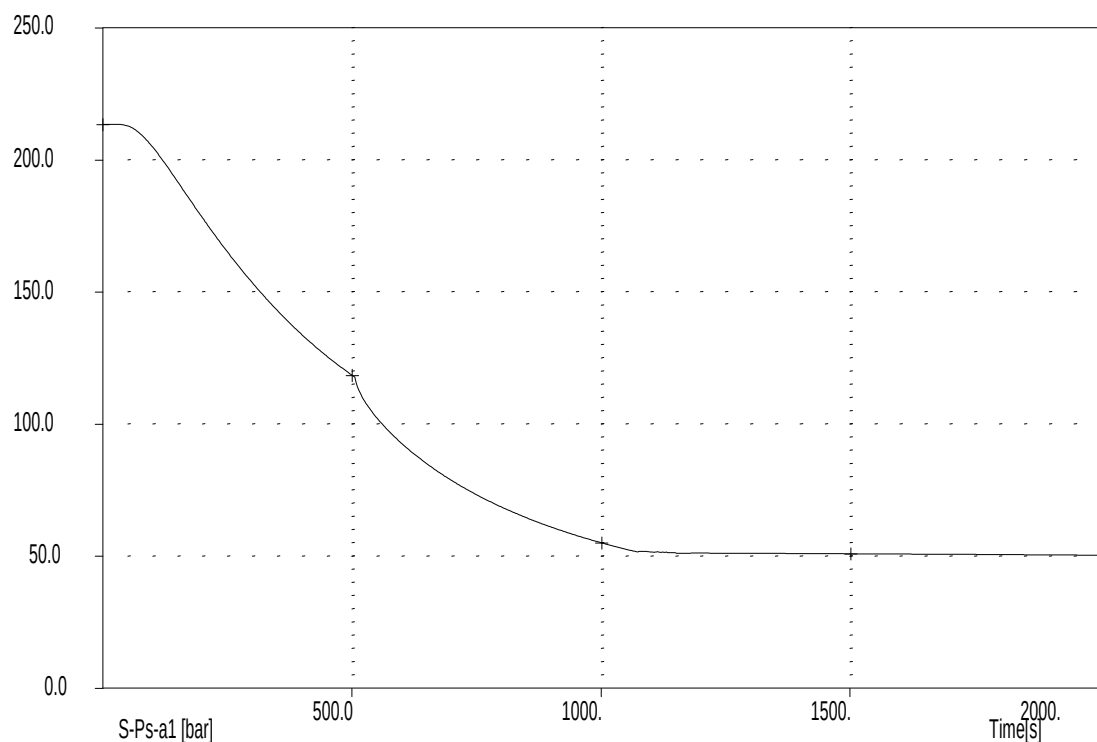


Figure 7-16 SCM supply pressure during bleed-down

Legend: S-Ps-a1; SCM supply pressure (bar)